08 Visual Encoding

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http://info-design-lab.github.io

Visualization helps humans solve analytical problems quickly and accurately

Over-the-Counter

National Market System

The companies listed below reflect the volume in 100's of shares on a daily basis and the closing price and net change are reflected from the previous day based on trades as provided under the NASD National Market System.

Stock	Sales	Close	Chg.
— A —			
[illegible]	[illegible]	[illegible]	[illegible]
— B —			
[illegible]	[illegible]	[illegible]	[illegible]
— D —			
[illegible]	[illegible]	[illegible]	[illegible]
— F —			
[illegible]	[illegible]	[illegible]	[illegible]
— H —			
[illegible]	[illegible]	[illegible]	[illegible]
— J-K —			
[illegible]	[illegible]	[illegible]	[illegible]
— O-P —			
[illegible]	[illegible]	[illegible]	[illegible]

57.46
7.48
17.58
66.40
66.34
63.34
34.70
0.86
40.98
15.01
78.49
91.80
31.99
74.34
20.57
58.33
7.63
17.97
66.87
68.14
64.42
34.84
0.86
41.76
15.03
78.49
93.27
32.24
75.09
21.03
-0.91%
1.49%
2.21%
-2.19%
0.48%
-1.60%
-0.82%
-0.12%
0.28%
-5.47%
-0.81%
0.79%
-0.51%
0.48%
0.48%
£ 7.52
£ 17.58
£ 68.37
£ 67.82
£ 65.47
£ 35.13
£ 0.87
£ 41.65
£ 15.90
79.13
£ 20.92
£ 104.2

28 day summary with change over previous period

Tweets	Tweet impressions	Profile visits	Mentions	Followers
66 ↓21.4%	1.4M ↑5.1%	21.9K ↓9.1%	447 ↓9.3%	56.8K ↑1,143

October 2015 • 3 days so far...

glucose 6.6

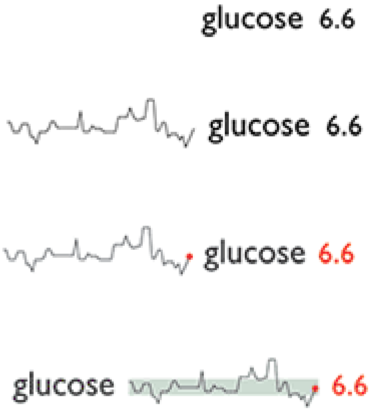
glucose 6.6

 glucose 6.6

glucose 6.6

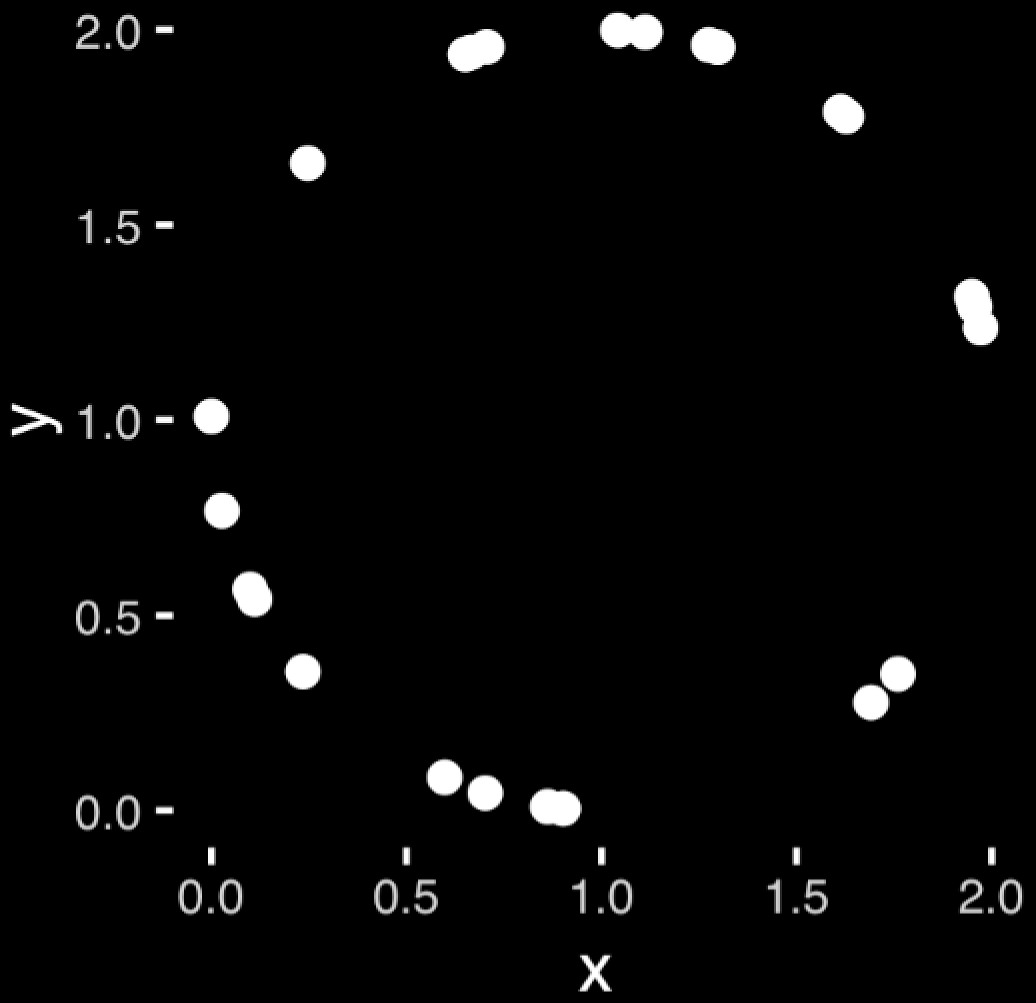

glucose 6.6

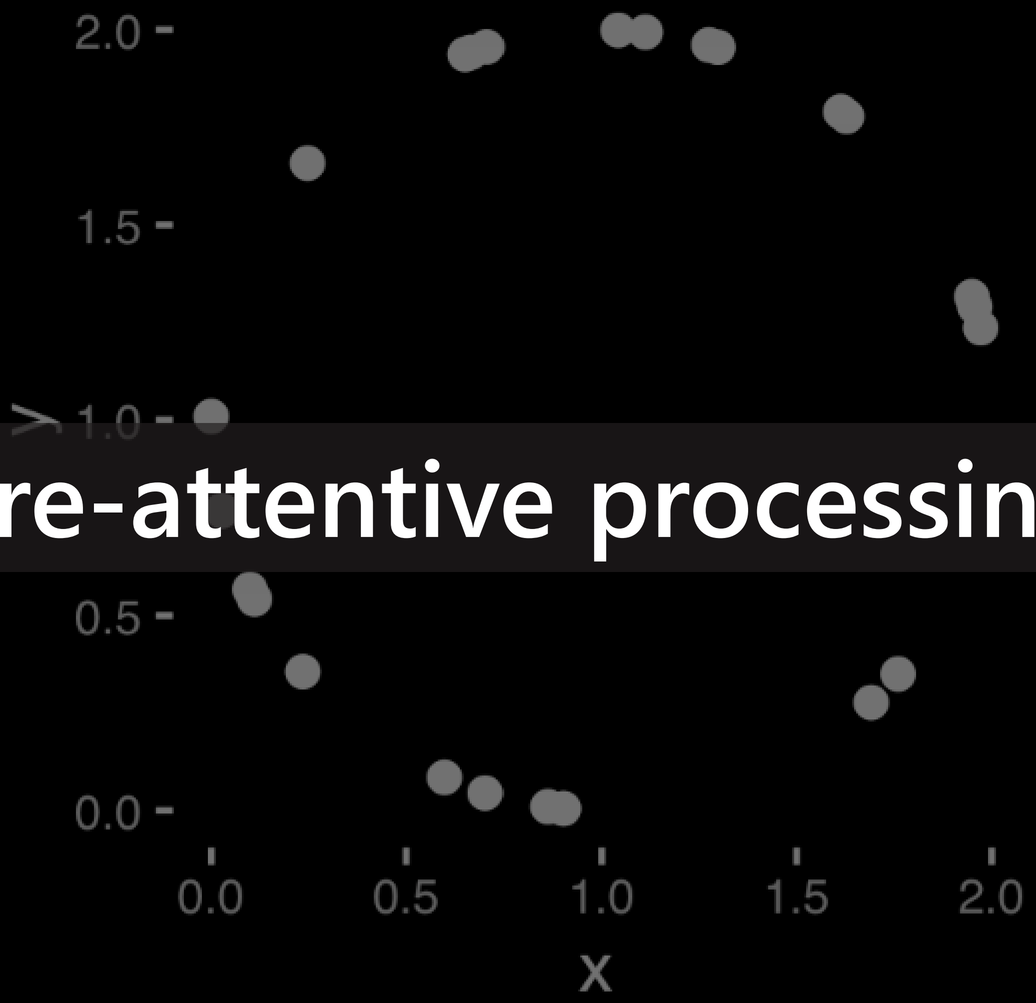

glucose 6.6


glucose 6.6
glucose 6.6
glucose 6.6
glucose 6.6

x	y
1.972	1.236
1.112	1.994
0.000	1.009
0.665	1.942
0.235	0.356
0.247	1.658
1.275	1.961
0.702	0.045
1.760	0.350
1.691	0.277
1.628	1.778
1.957	1.290

x	y
0.111	0.542
0.902	0.005
0.598	0.085
1.613	1.790
1.298	1.955
0.651	1.937
1.949	1.316
0.099	0.567
0.862	0.010
0.027	0.768
0.706	1.956
1.042	1.999


2.0
1.5
1.0
0.5
0.0
y
0.0
0.5
1.0
1.5
2.0
x



pre-attentive processing

141692653289793238462643383279
302884197169399375104820974944
792307816406286208998628053482
342117067982148086513282306647

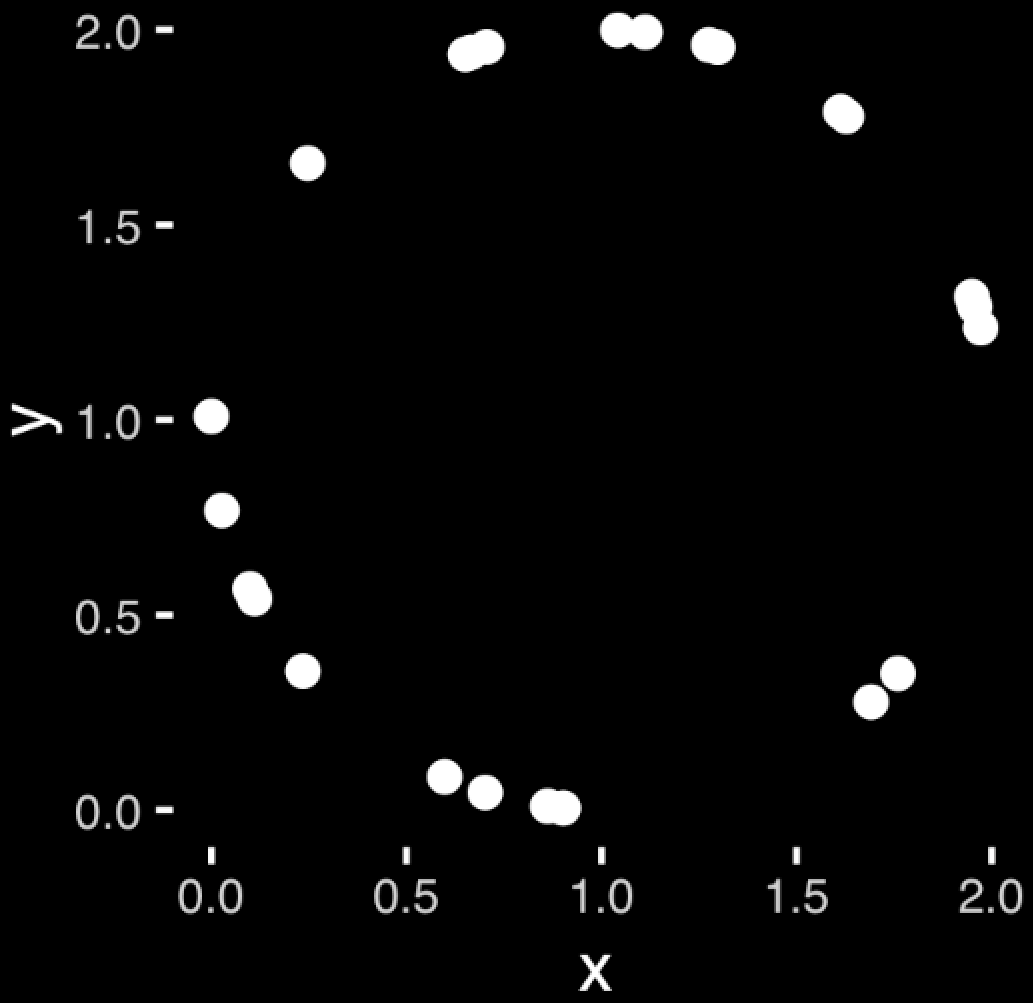
14169 26532 89793 23846 26433 83279
30288 41971 69399 37510 48209 74944
79230 78164 06286 20899 86280 53482
34211 70679 82148 08651 32823 06647

141692653289793238462643383279
302884197169399375104820974944
792307816406286208998628053482
342117067982148086513282306647

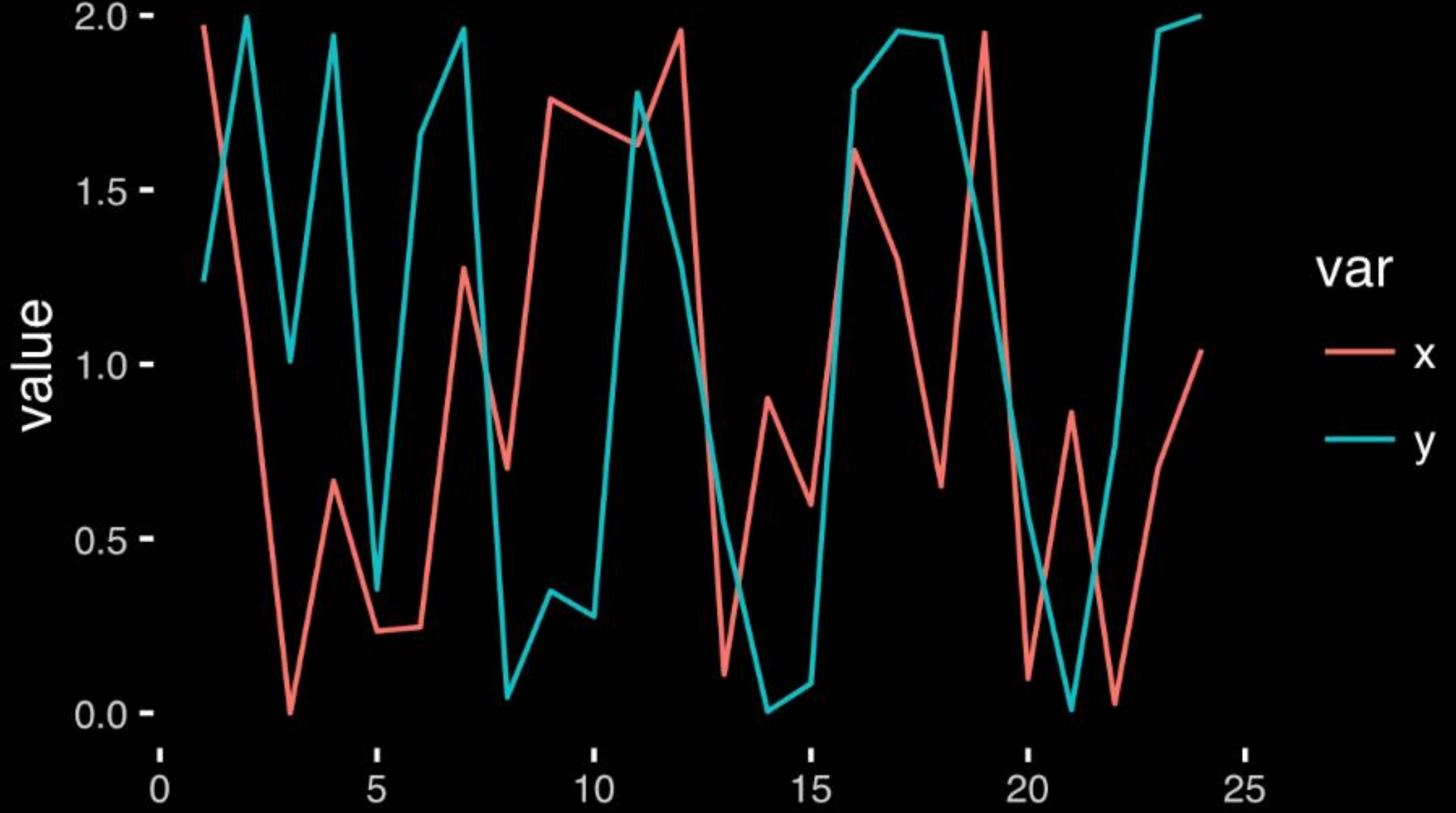
A graph is an encoding of the data

x	y
1.972	1.236
1.112	1.994
0.000	1.009
0.665	1.942
0.235	0.356
0.247	1.658
1.275	1.961
0.702	0.045
1.760	0.350
1.691	0.277
1.628	1.778
1.957	1.290

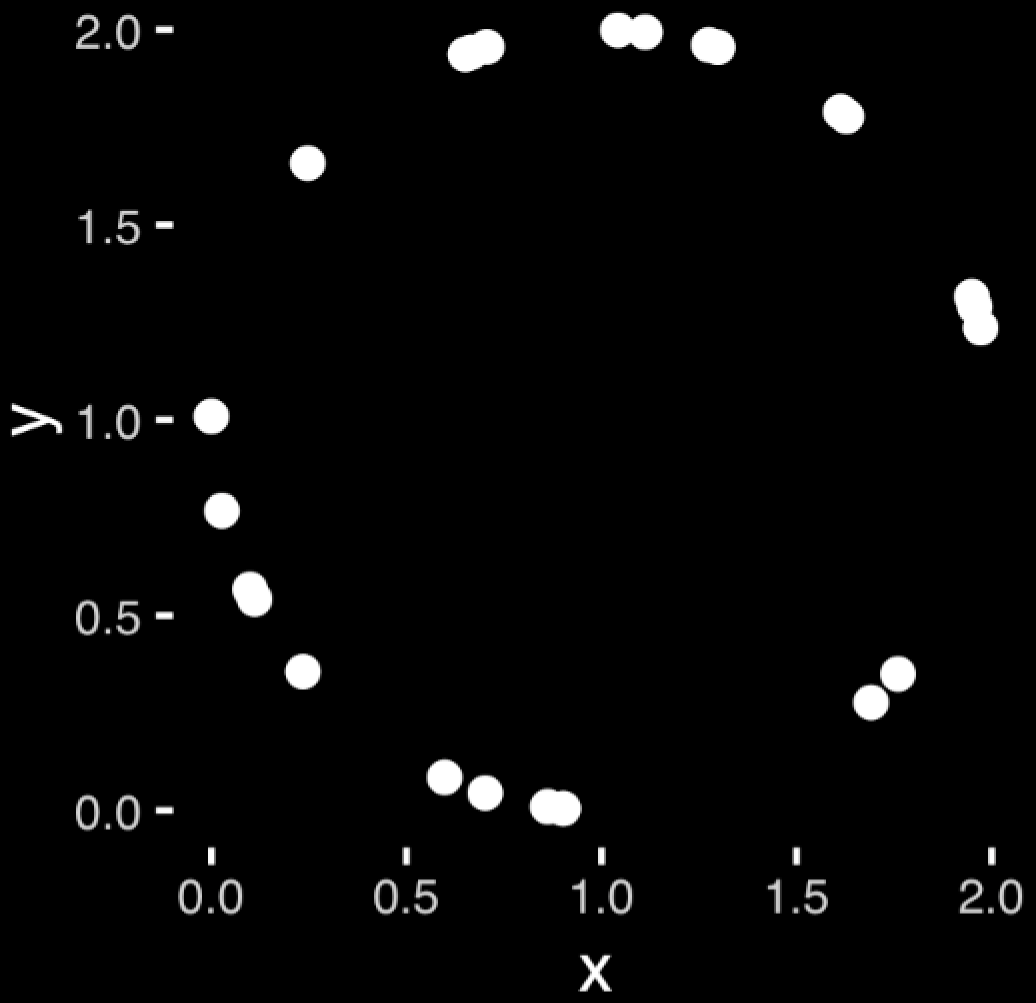
x	y
0.111	0.542
0.902	0.005
0.598	0.085
1.613	1.790
1.298	1.955
0.651	1.937
1.949	1.316
0.099	0.567
0.862	0.010
0.027	0.768
0.706	1.956
1.042	1.999


2.0
1.5
1.0
0.5
0.0
y
0.0
0.5
1.0
1.5
2.0
x

n	x	y	n	x	y
1	1.972	1.236	13	0.111	0.542
2	1.112	1.994	14	0.902	0.005
3	0.000	1.009	15	0.598	0.085
4	0.665	1.942	16	1.613	1.790
5	0.235	0.356	17	1.298	1.955
6	0.247	1.658	18	0.651	1.937
7	1.275	1.961	19	1.949	1.316
8	0.702	0.045	20	0.099	0.567
9	1.760	0.350	21	0.862	0.010
10	1.691	0.277	22	0.027	0.768
11	1.628	1.778	23	0.706	1.956
12	1.957	1.290	24	1.042	1.999

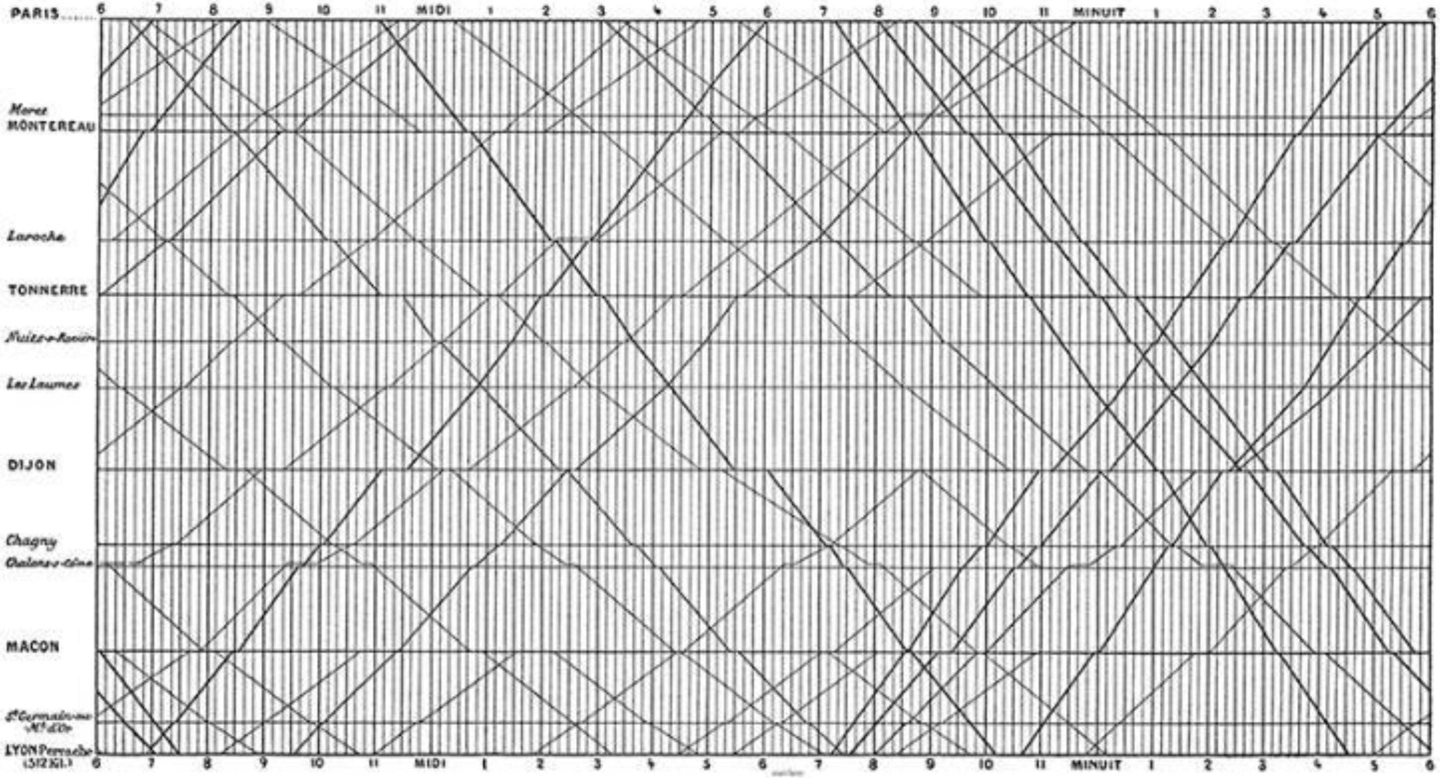


value
2.0
1.5
1.0
0.5
0.0
0
5
10
15
20
25
var
x
y



y
2.0
1.5
1.0
0.5
0.0
0.0
0.5
1.0
1.5
2.0
x

Good visualizations optimize for the human visual system



E. J. Marey, *La méthode graphique* (Paris, 1885), 20. The method is attributed to the French engineer, Ibry.

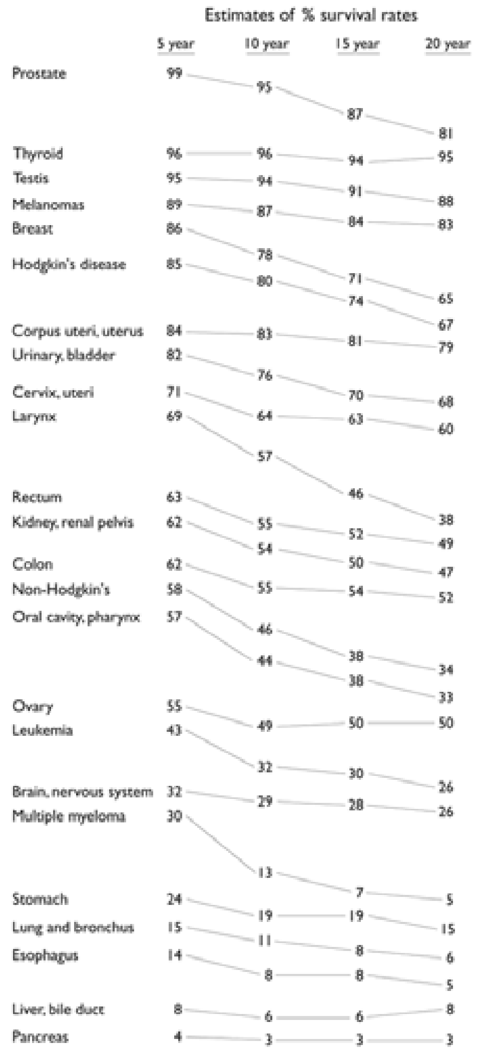
Estimates of relative survival rates, by cancer site

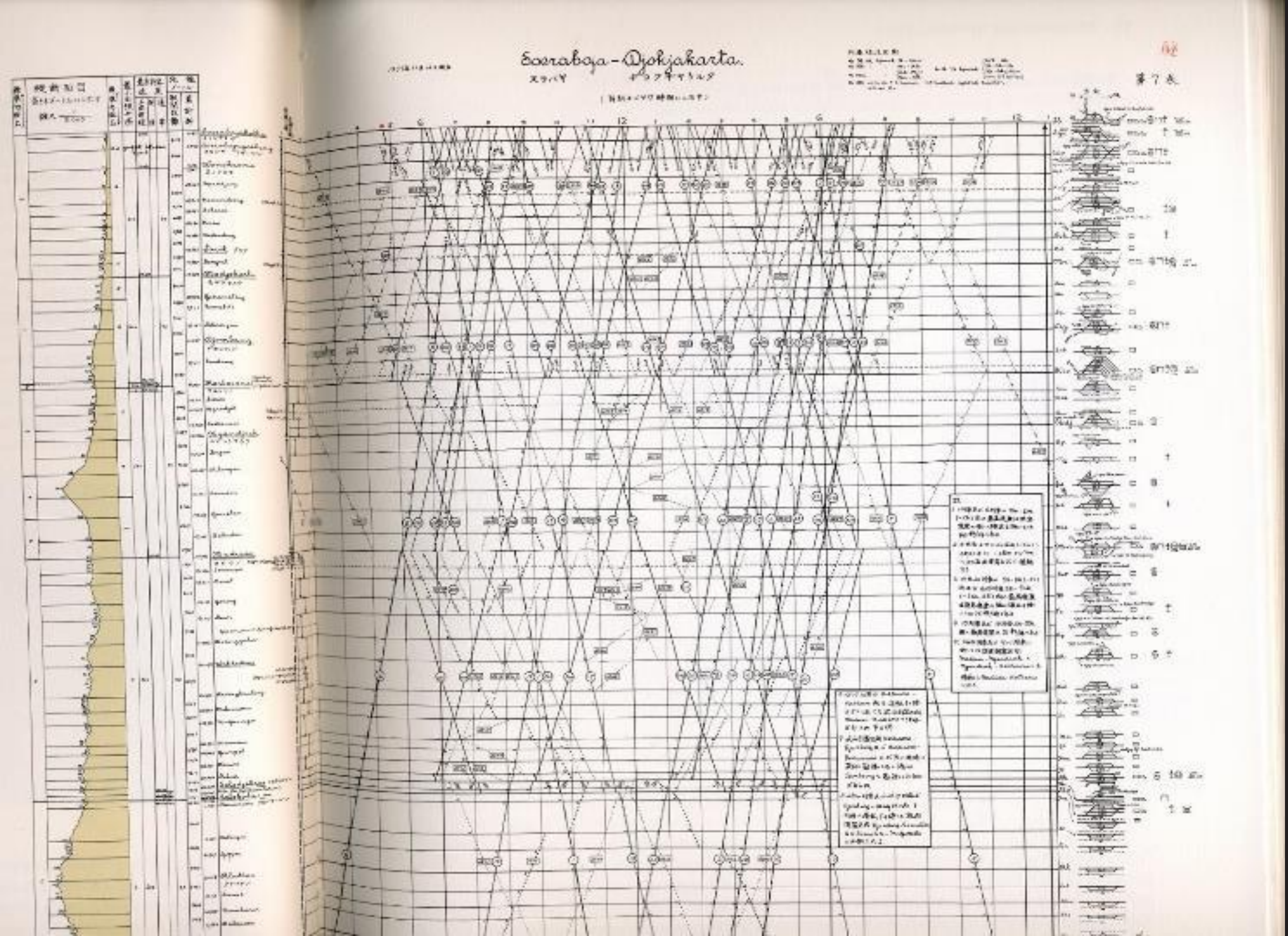
% survival rates and their standard errors

	5 year		10 year		15 year		20 year	
Prostate	98.8	0.4	95.2	0.9	87.1	1.7	81.1	3.0
Thyroid	96.0	0.8	95.8	1.2	94.0	1.6	95.4	2.1
Testis	94.7	1.1	94.0	1.3	91.1	1.8	88.2	2.3
Melanomas	89.0	0.8	86.7	1.1	83.5	1.5	82.8	1.9
Breast	86.4	0.4	78.3	0.6	71.3	0.7	65.0	1.0
Hodgkin's disease	85.1	1.7	79.8	2.0	73.8	2.4	67.1	2.8
Corpus uteri, uterus	84.3	1.0	83.2	1.3	80.8	1.7	79.2	2.0
Urinary, bladder	82.1	1.0	76.2	1.4	70.3	1.9	67.9	2.4
Cervix, uteri	70.5	1.6	64.1	1.8	62.8	2.1	60.0	2.4
Larynx	68.8	2.1	56.7	2.5	45.8	2.8	37.8	3.1
Rectum	62.6	1.2	55.2	1.4	51.8	1.8	49.2	2.3
Kidney, renal pelvis	61.8	1.3	54.4	1.6	49.8	2.0	47.3	2.6
Colon	61.7	0.8	55.4	1.0	53.9	1.2	52.3	1.6
Non-Hodgkin's	57.8	1.0	46.3	1.2	38.3	1.4	34.3	1.7
Oral cavity, pharynx	56.7	1.3	44.2	1.4	37.5	1.6	33.0	1.8
Ovary	55.0	1.3	49.3	1.6	49.9	1.9	49.6	2.4
Leukemia	42.5	1.2	32.4	1.3	29.7	1.5	26.2	1.7
Brain, nervous system	32.0	1.4	29.2	1.5	27.6	1.6	26.1	1.9
Multiple myeloma	29.5	1.6	12.7	1.5	7.0	1.3	4.8	1.5
Stomach	23.8	1.3	19.4	1.4	19.0	1.7	14.9	1.9
Lung and bronchus	15.0	0.4	10.6	0.4	8.1	0.4	6.5	0.4
Esophagus	14.2	1.4	7.9	1.3	7.7	1.6	5.4	2.0
Liver, bile duct	7.5	1.1	5.8	1.2	6.3	1.5	7.6	2.0
Pancreas	4.0	0.5	3.0	1.5	2.7	0.6	2.7	0.8

Estimates of relative survival rates, by cancer site

	% survival rates and their standard errors							
	5 year		10 year		15 year		20 year	
Prostate	98.8	0.4	95.2	0.9	87.1	1.7	81.1	3.0
Thyroid	96.0	0.8	95.8	1.2	94.0	1.6	95.4	2.1
Testis	94.7	1.1	94.0	1.3	91.1	1.8	88.2	2.3
Melanomas	89.0	0.8	86.7	1.1	83.5	1.5	82.8	1.9
Breast	86.4	0.4	78.3	0.6	71.3	0.7	65.0	1.0
Hodgkin's disease	85.1	1.7	79.8	2.0	73.8	2.4	67.1	2.8
Corpus uteri, uterus	84.3	1.0	83.2	1.3	80.8	1.7	79.2	2.0
Urinary, bladder	82.1	1.0	76.2	1.4	70.3	1.9	67.9	2.4
Cervix, uteri	70.5	1.6	64.1	1.8	62.8	2.1	60.0	2.4
Larynx	68.8	2.1	56.7	2.5	45.8	2.8	37.8	3.1
Rectum	62.6	1.2	55.2	1.4	51.8	1.8	49.2	2.3
Kidney, renal pelvis	61.8	1.3	54.4	1.6	49.8	2.0	47.3	2.6
Colon	61.7	0.8	55.4	1.0	53.9	1.2	52.3	1.6
Non-Hodgkin's	57.8	1.0	46.3	1.2	38.3	1.4	34.3	1.7
Oral cavity, pharynx	56.7	1.3	44.2	1.4	37.5	1.6	33.0	1.8
Ovary	55.0	1.3	49.3	1.6	49.9	1.9	49.6	2.4
Leukemia	42.5	1.2	32.4	1.3	29.7	1.5	26.2	1.7
Brain, nervous system	32.0	1.4	29.2	1.5	27.6	1.6	26.1	1.9
Multiple myeloma	29.5	1.6	12.7	1.5	7.0	1.3	4.8	1.5
Stomach	23.8	1.3	19.4	1.4	19.0	1.7	14.9	1.9
Lung and bronchus	15.0	0.4	10.6	0.4	8.1	0.4	6.5	0.4
Esophagus	14.2	1.4	7.9	1.3	7.7	1.6	5.4	2.0
Liver, bile duct	7.5	1.1	5.8	1.2	6.3	1.5	7.6	2.0
Pancreas	4.0	0.5	3.0	1.5	2.7	0.6	2.7	0.8





Timetable for Java Railroad line, 1937. In Tufte 1990 standard slope ‰

cross section
height in meters
1/3000

standard slope ‰

minimum radius

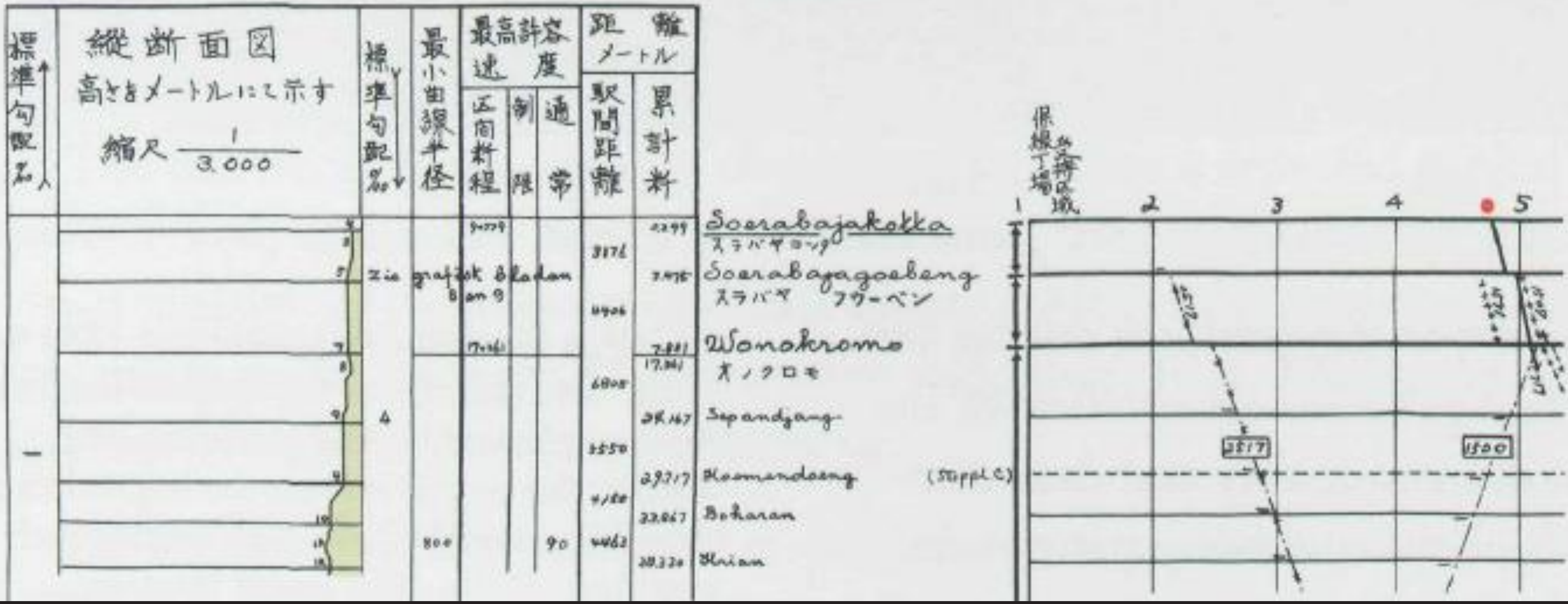
distance

maximum speed

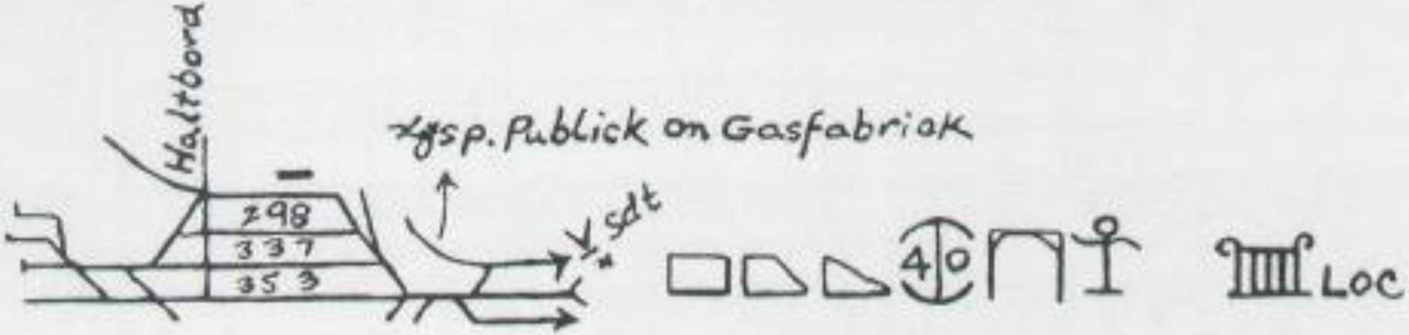
normal speed

distances between
stations and
cumulative distance

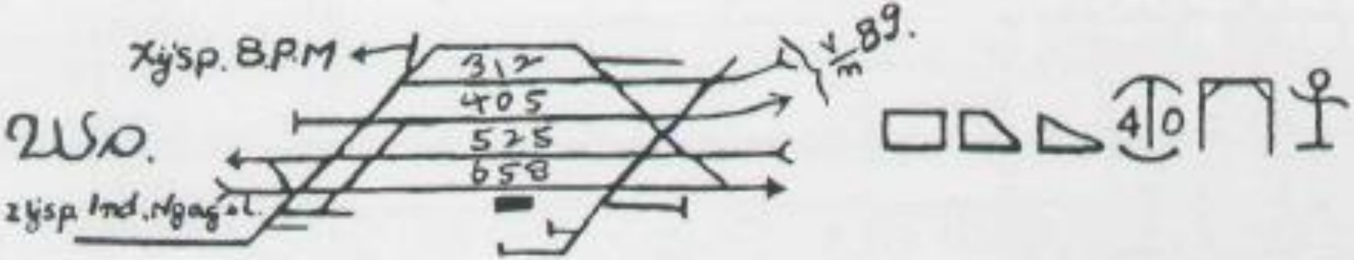
town names
major stops
underlined



Sb.


Haltbord
Xjsp. Publick on Gasfabriak
sdt
298
337
353
40
Loc

Wo.


Xjsp. B.P.M
Zjsp. Ind. Ngagel.
89.
312
405
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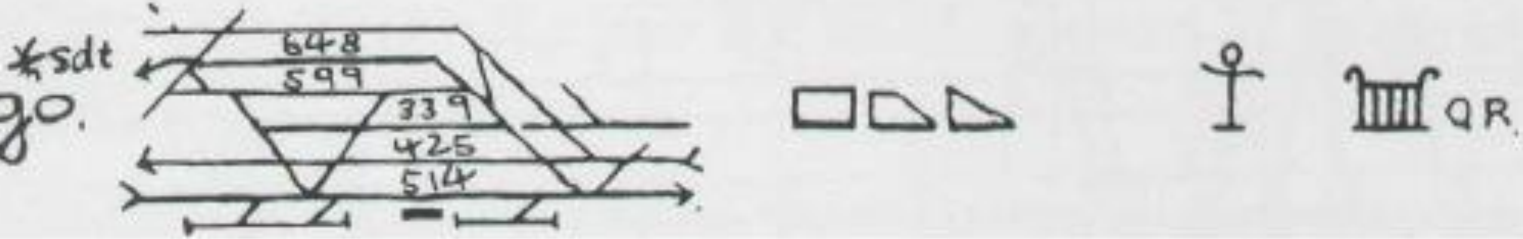
658
40

Sgo.


sdt
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339
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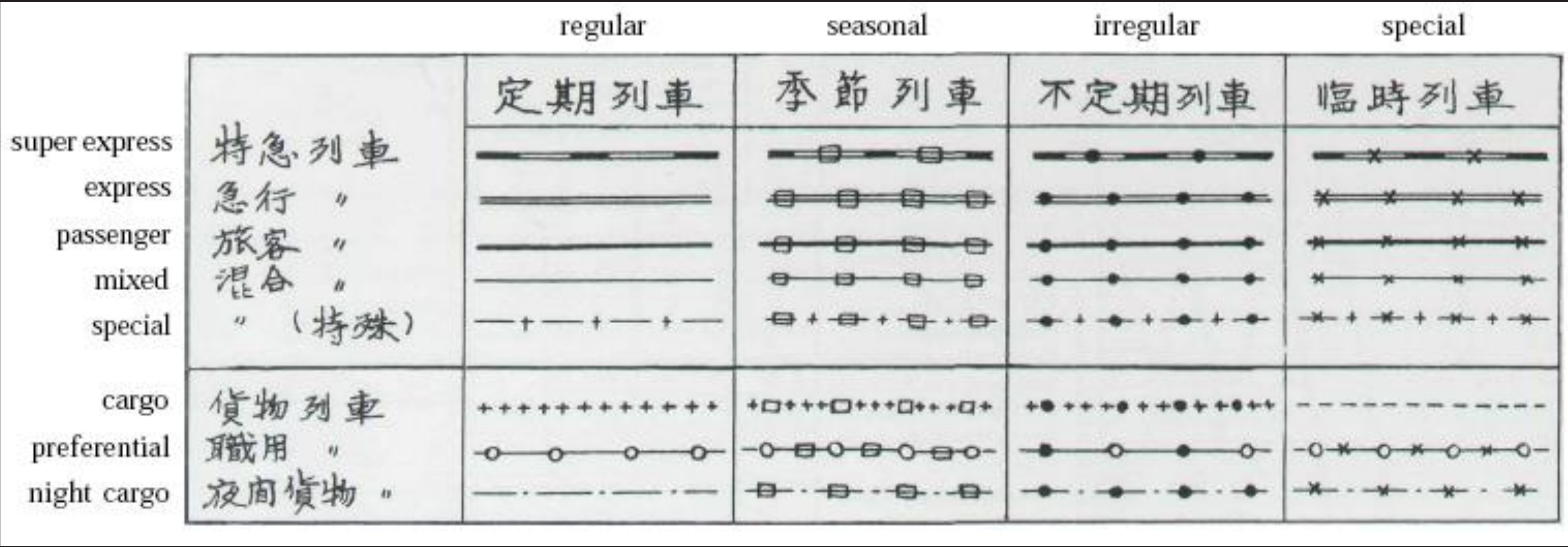
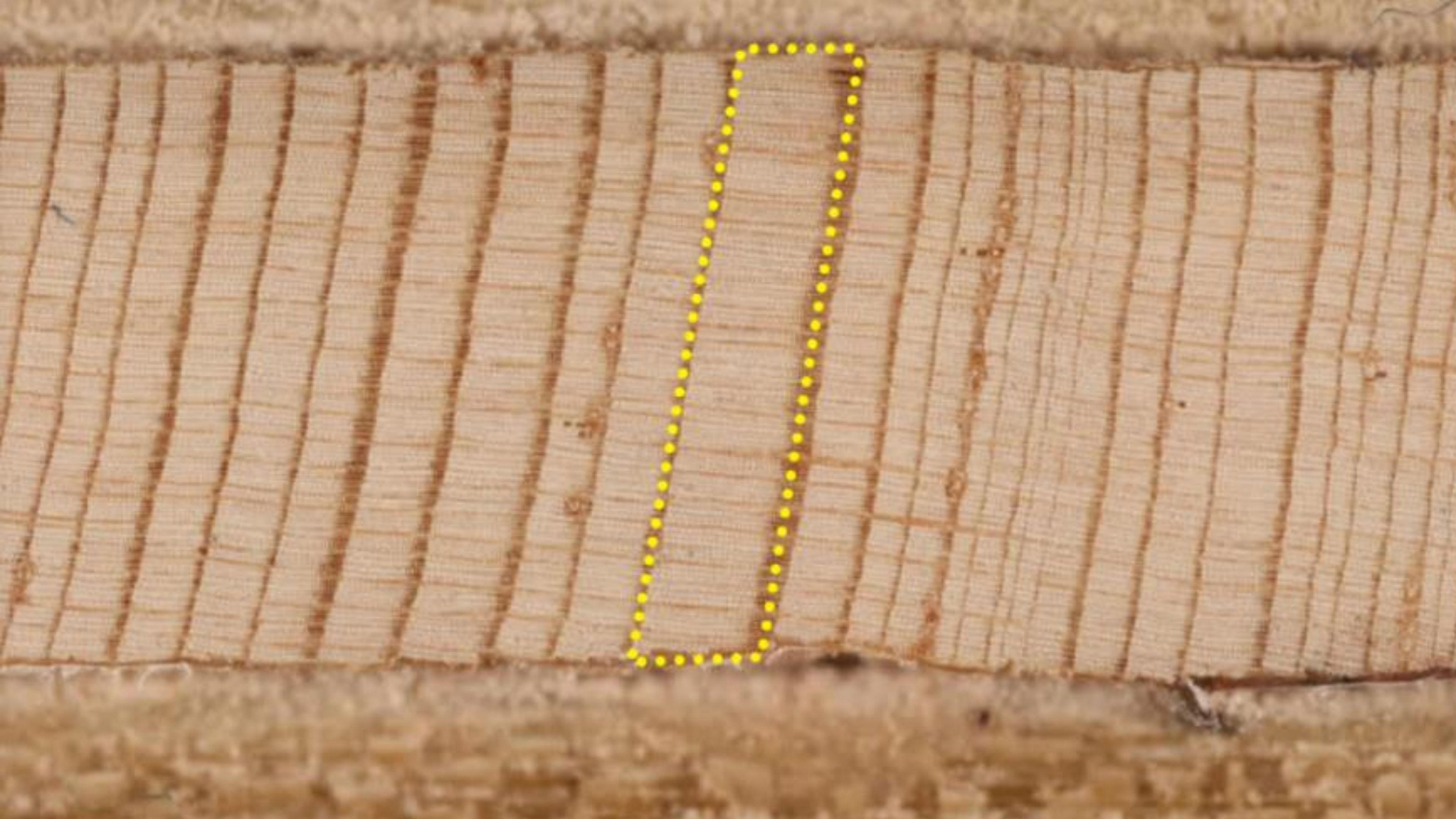
514
QR.


regular
seasonal
irregular
special
定期列車
季節列車
不定期列車
臨時列車
super express
特急列車
express
急行 〃
passenger
旅客 〃
mixed
混合 〃
special
〃 (特殊)
cargo
貨物列車
preferential
職用 〃
night cargo
夜间貨物 〃




5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
80
85
90
95
100
105
110
115
120
125
130


12
1
2
3
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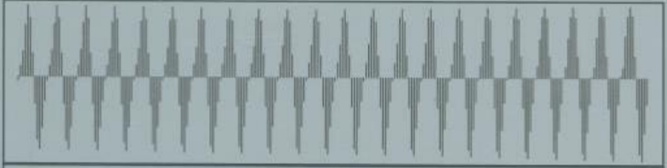
11

How do the human visual & cognitive systems work?

How do humans decode a graph?



The Elements of Graphing Data

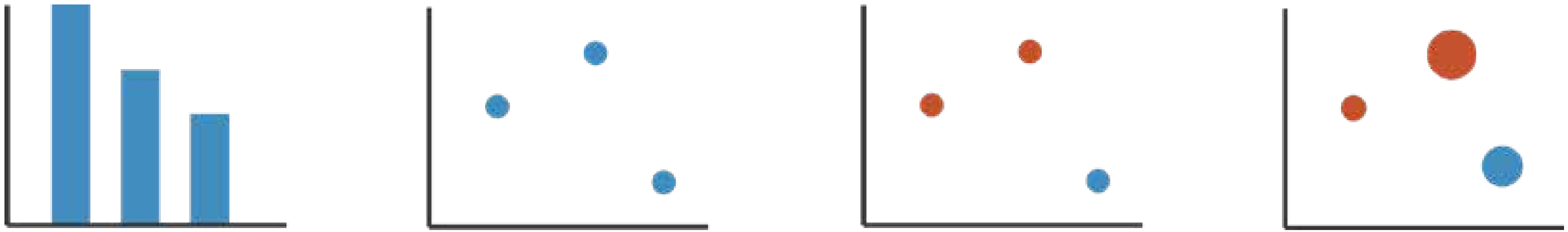
William S. Cleveland

Cleveland’s three visual operations of pattern perception:

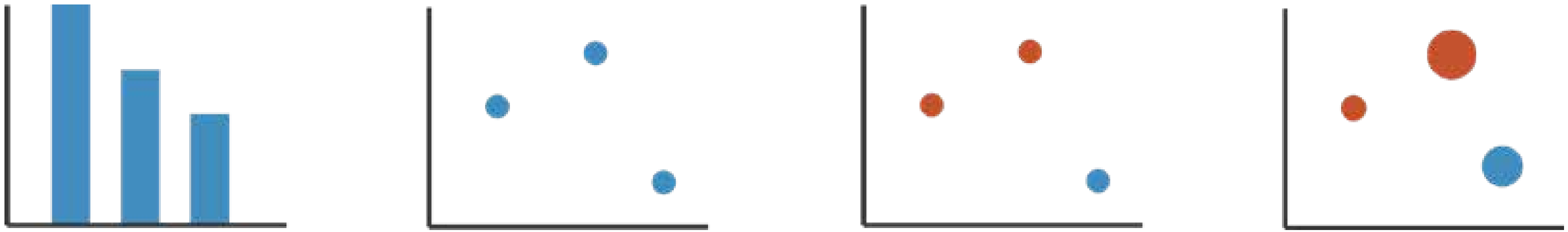
1. Detection
2. Assembly
3. Estimation

1. Detection

- Detection is the operation of recognizing that a geometric object encodes a physical value



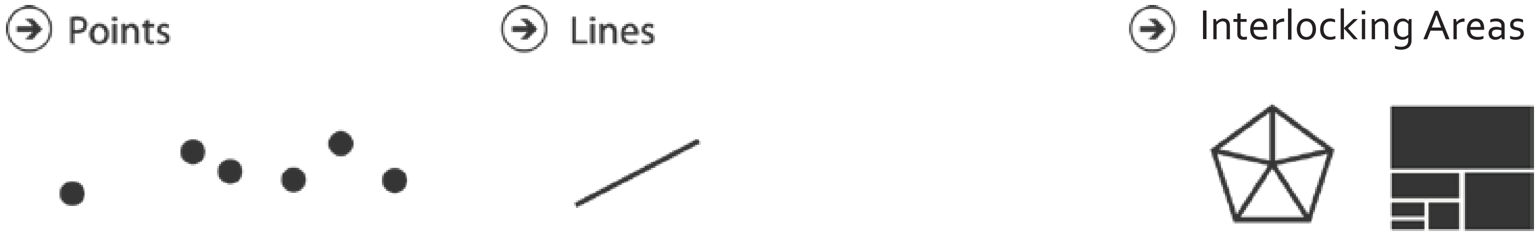
- Detection is the operation of recognizing that a geometric object encodes a physical value



- marks & channels
  - marks: represent items or links
  - channels: change appearance of marks based on attributes

Marks for items

- basic geometric elements

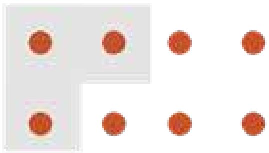


0D 1D 2D

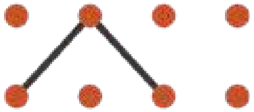
- 3D mark: volume, rarely used

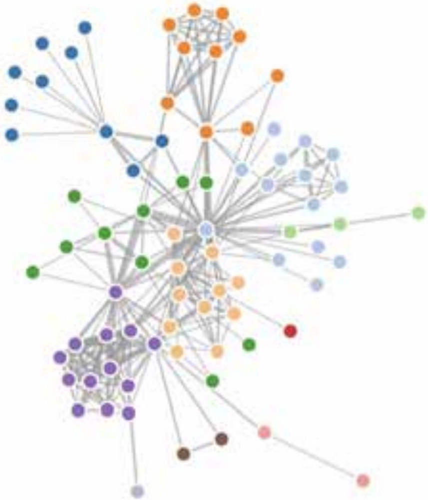
Marks for links

Containment

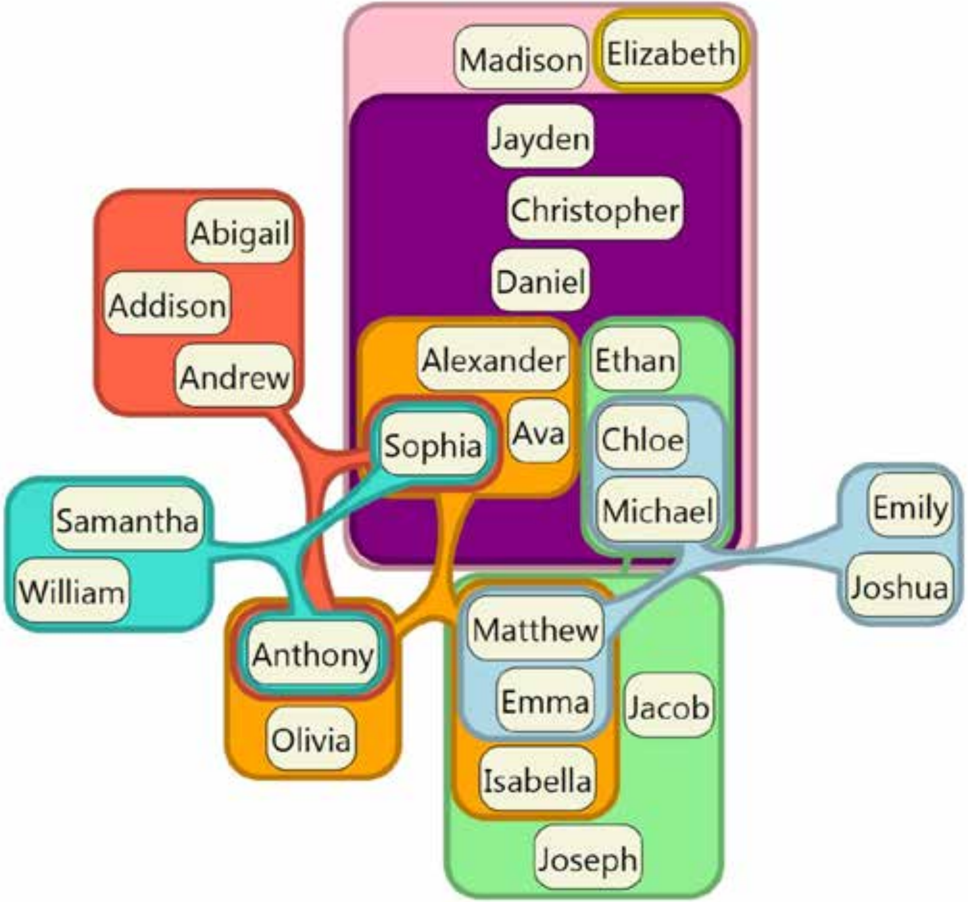


Connection





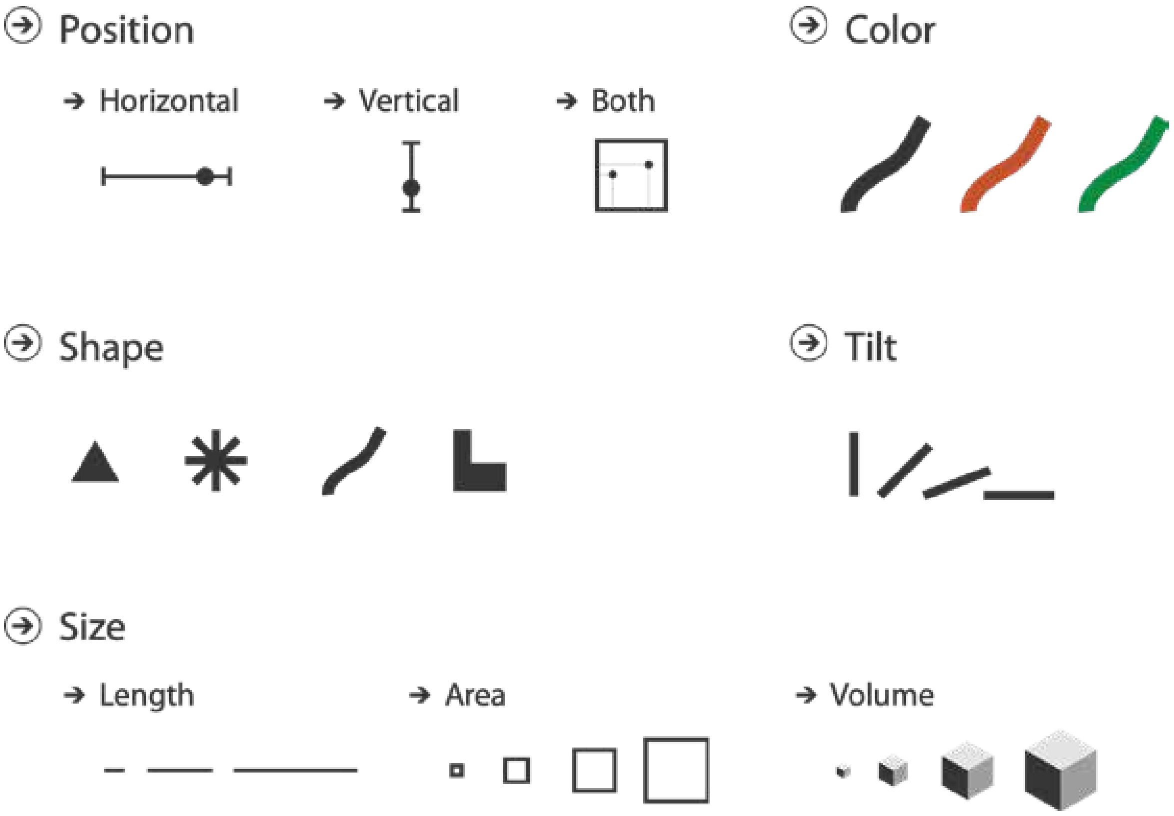
Containment can be nested



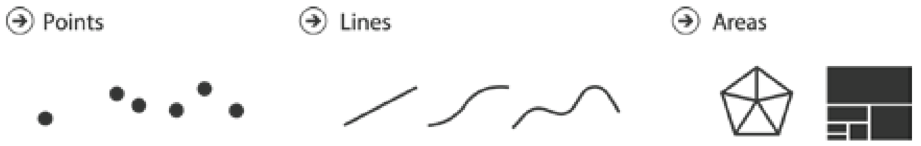
Channels

- control appearance of marks
  - proportional to or based on attributes

- many names
  - **visual channels**
  - visual variables
  - retinal channels
  - visual dimensions
  - ...



Definitions: Marks and channels

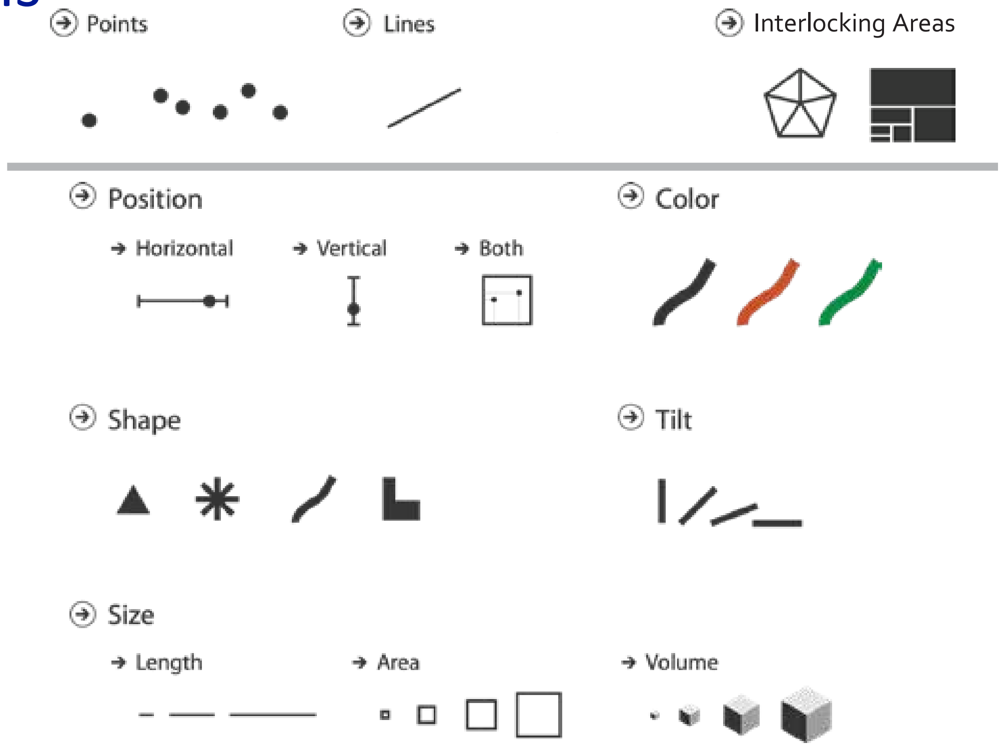


- marks
  - geometric primitives

Definitions: Marks and channels

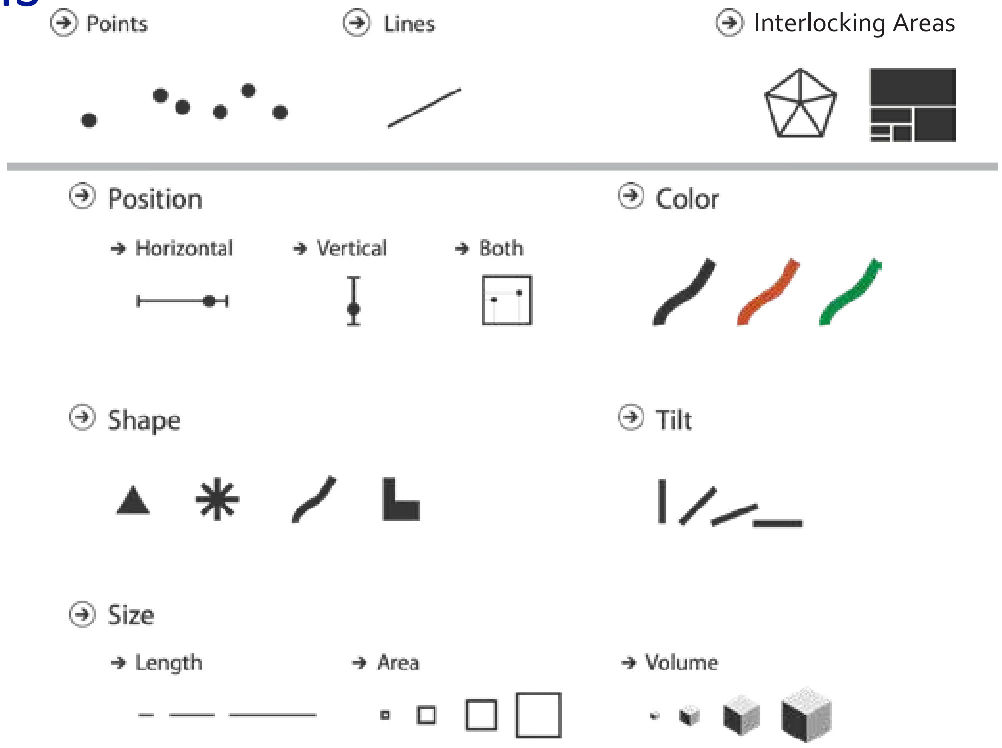
- marks
  - geometric primitives

- channels
  - control appearance of marks



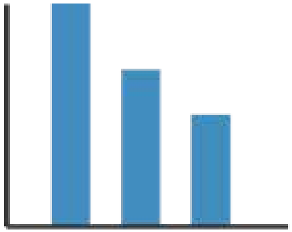
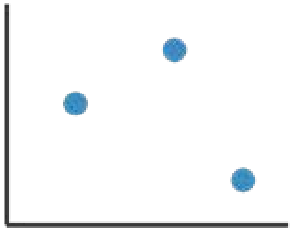
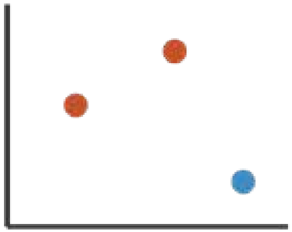
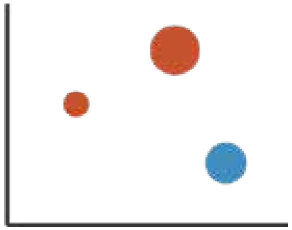
Definitions: Marks and channels

- marks
  - geometric primitives
- channels
  - control appearance of marks
- channel properties differ
  - type & amount of information that can be conveyed to human perceptual system

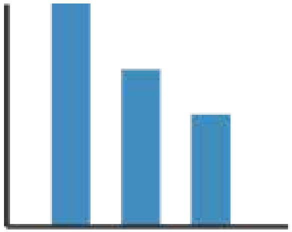


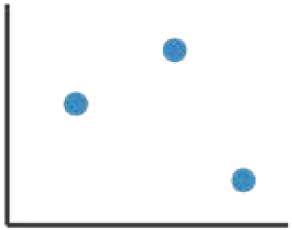
Visual encoding

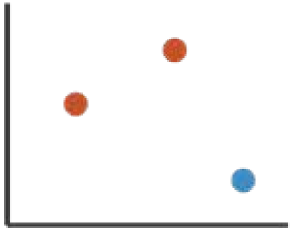
- analyze idiom structure as combination of marks and channels

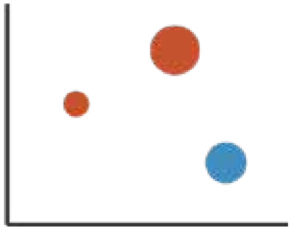





Visual encoding





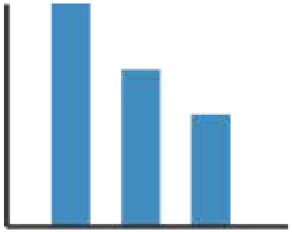


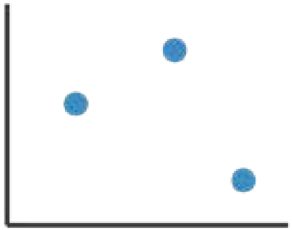


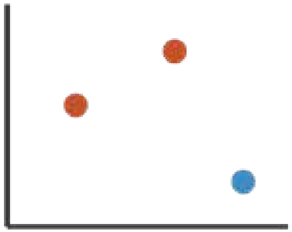
1:
vertical position

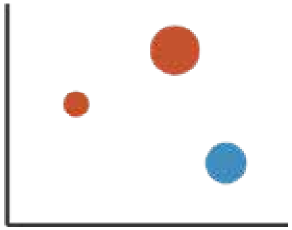
mark: line

Visual encoding









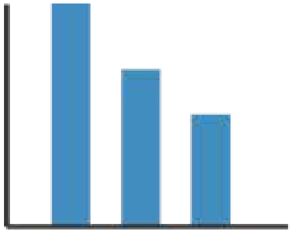
1:
vertical position

2:
vertical position
horizontal position

mark: line

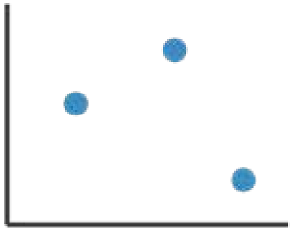
mark: line

Visual encoding



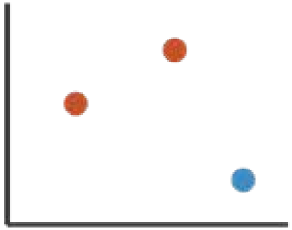
1:
vertical position

mark: line



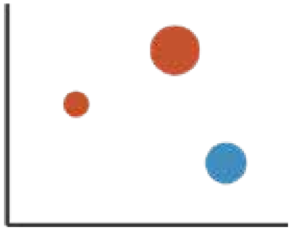
2:
vertical position
horizontal position

mark: line

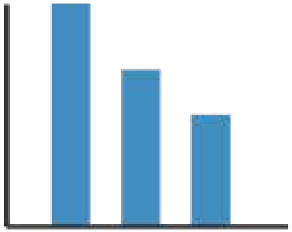


3:
vertical position
horizontal position
colour hue

mark: line

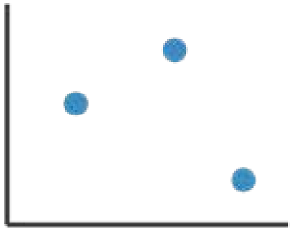


Visual encoding



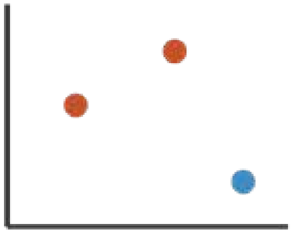
1:
vertical position

mark: line



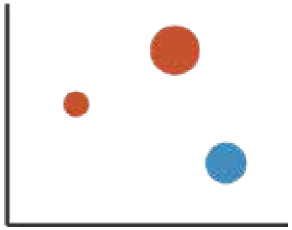
2:
vertical position
horizontal position

mark: line



3:
vertical position
horizontal position
colour hue

mark: line

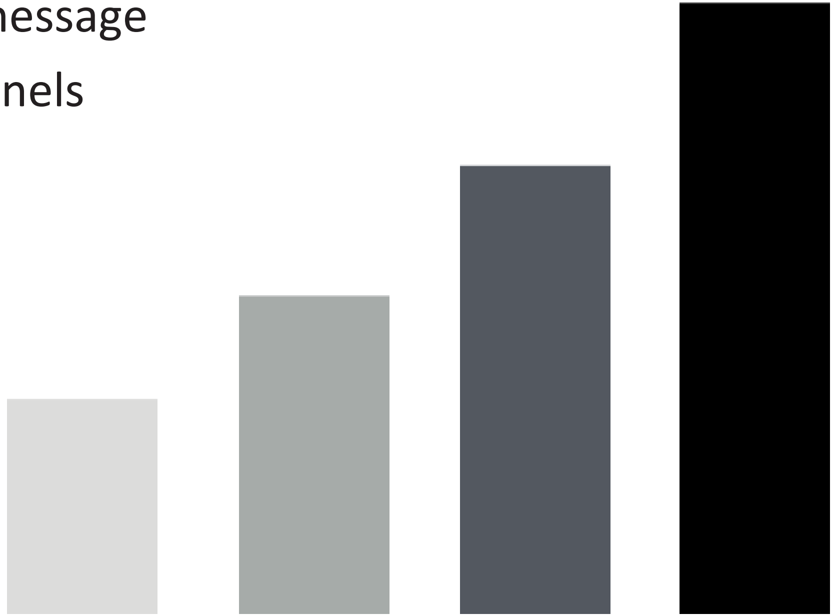


4:
vertical position
horizontal position
colour hue
size (area)

mark: line

Redundant encoding

- multiple channels
  - sends stronger message
  - but uses up channels



The Eye

- 70% of body's sense receptors reside in our eyes
- The eye and the visual cortex of the brain form a massively parallel processor that provides the highest-bandwidth channel into human cognitive centers.“
  — Colin Ware, Information Visualization, 2004
- Important to understand how visual perception works in order to effectively design visualizations

The Eye

- The eye is not a camera!

- Better metaphor for vision: “dynamic and ongoing construction project"
  — *Healey, 1995*

- Attention is selective (Filtering)

The Eye

○ **Cameras**
— Good optics
— Single focus, white balance, exposure
— "Full image capture"

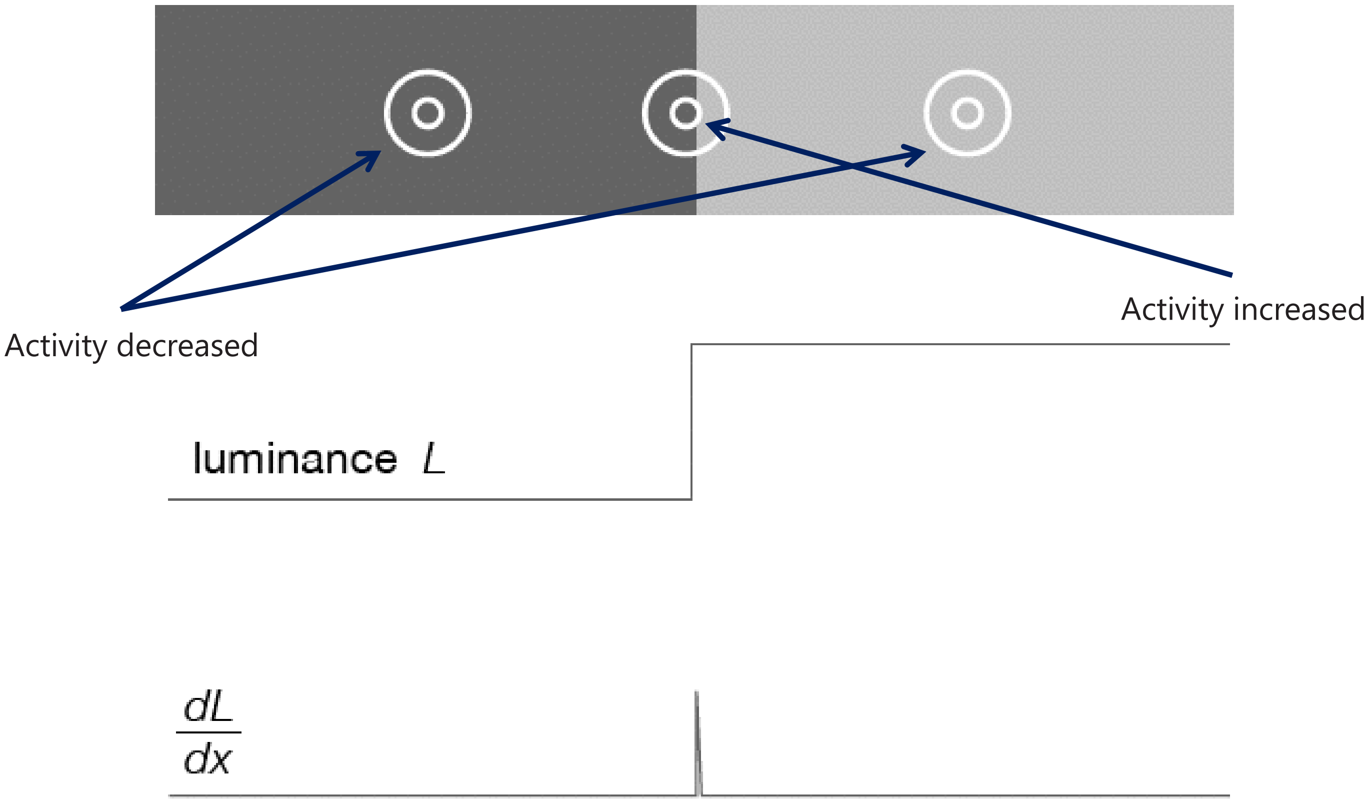
○ **Eyes**
— Relatively poor optics
— Constantly scanning (saccades)
— Constantly adjusting focus
— Constantly adapting
— Mental reconstruction of image

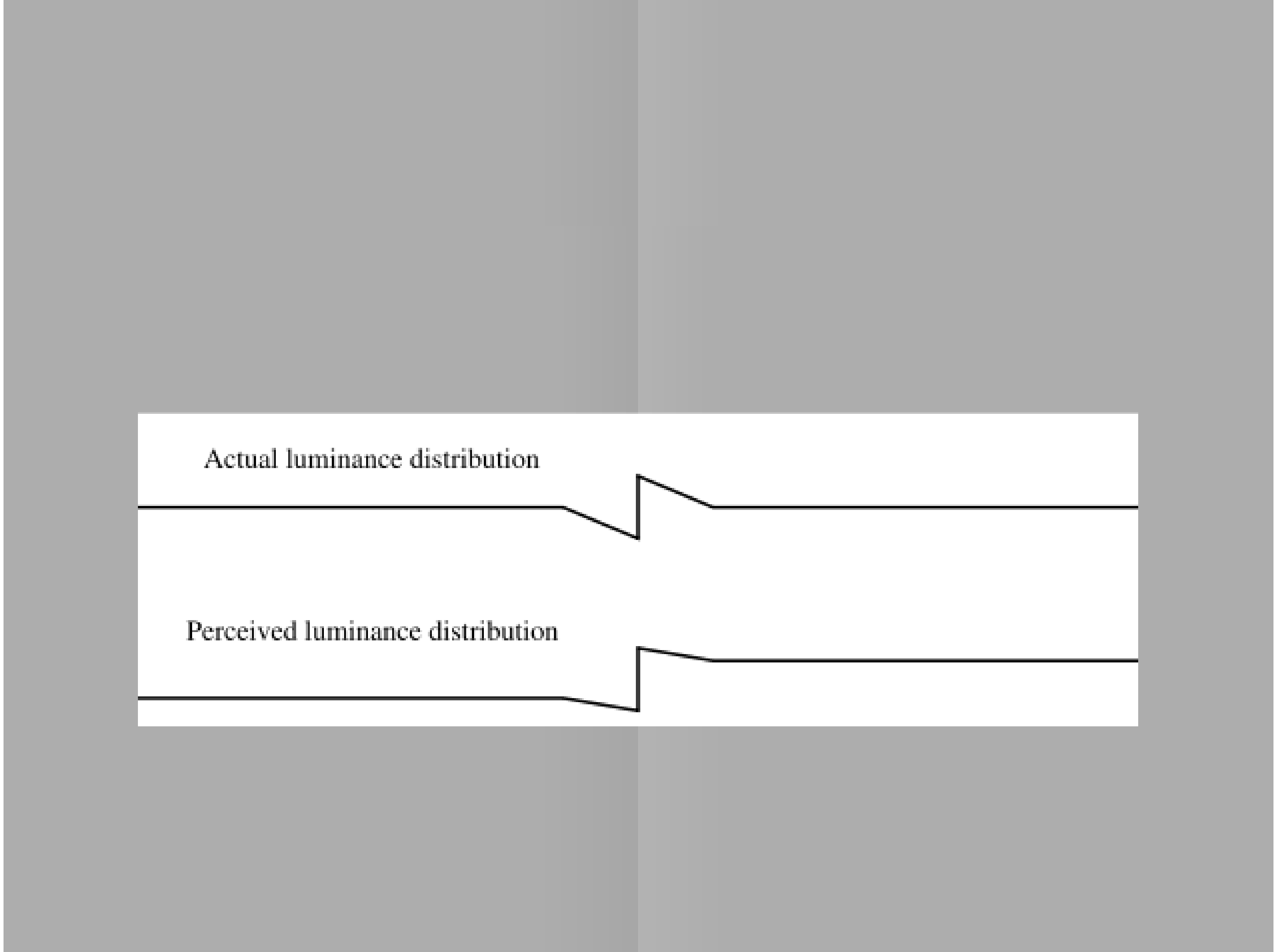
How the eye detects information

- Our visual system sees differences, not absolute values, and is attracted to edges.
- Maximize the contrast with the background if the outlines of shapes are important.
- Our visual system constructs surface colour based largely on edge contrast information.
- We have higher contrast sensitivity in the luminance than in the chrominance channel.
- Our visual system corrects (misreads) information based on perceived visual properties

The Eye – Edge detection

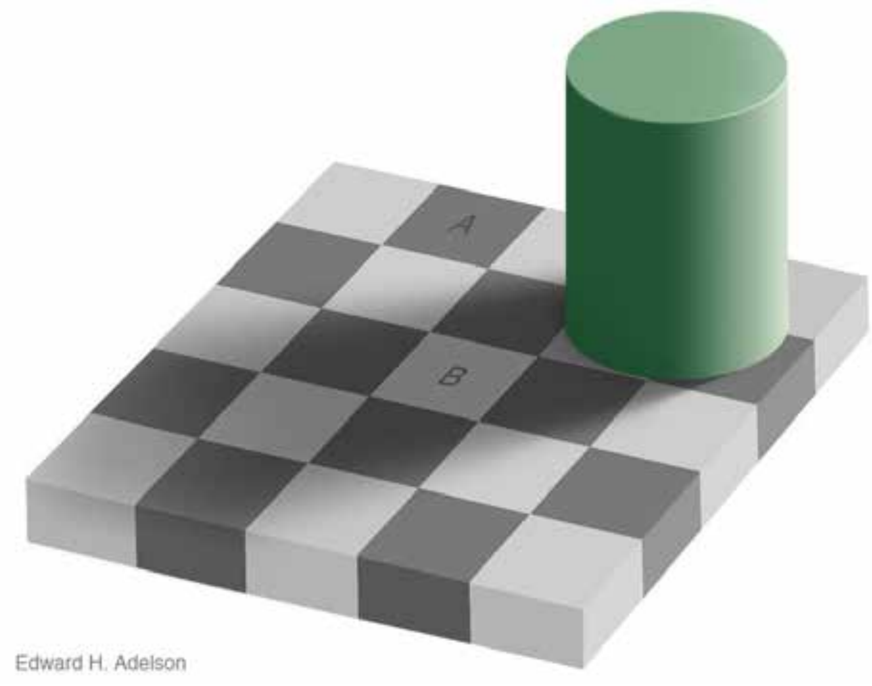
The Eye – Edge detection




Actual luminance distribution
Perceived luminance distribution

Relative luminance judgements

- perception of luminance is contextual based on contrast with surroundings



Relative luminance judgements

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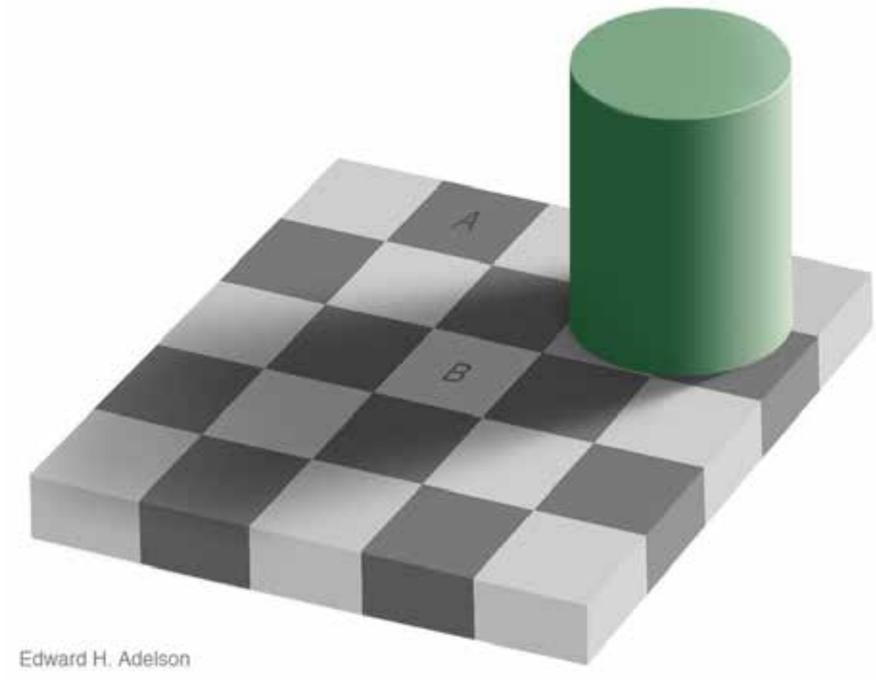
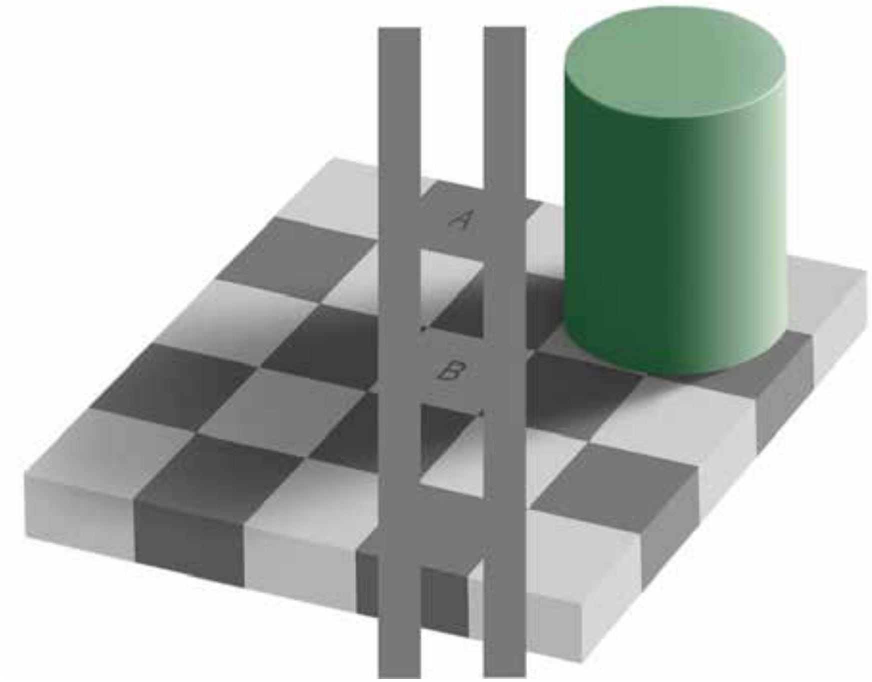
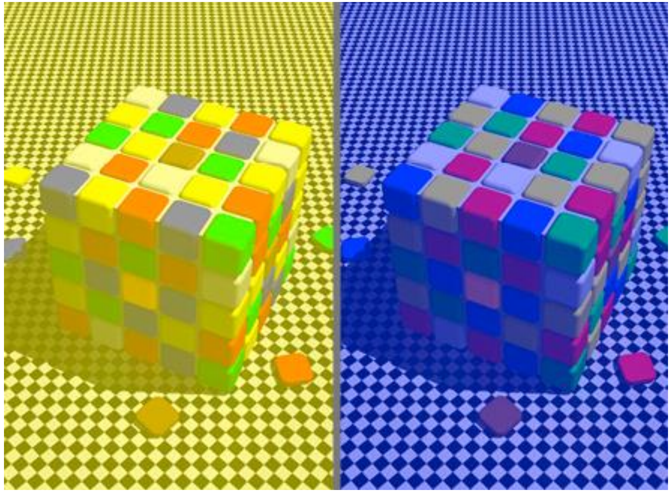




http://persci.mit.edu/gallery/checkershadow

Relative color judgements

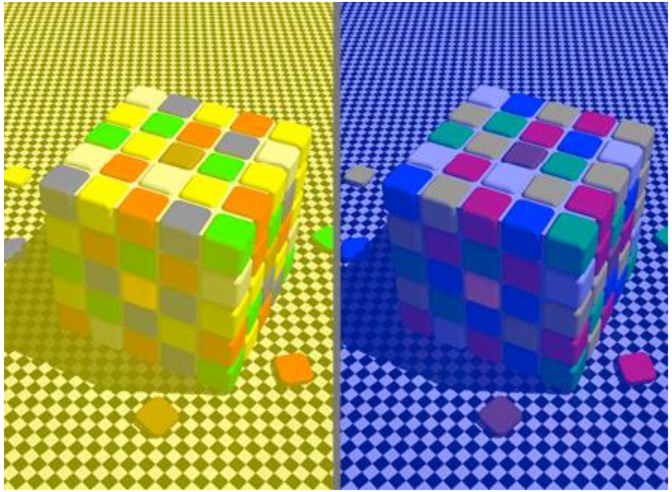
- color constancy across broad range of illumination conditions

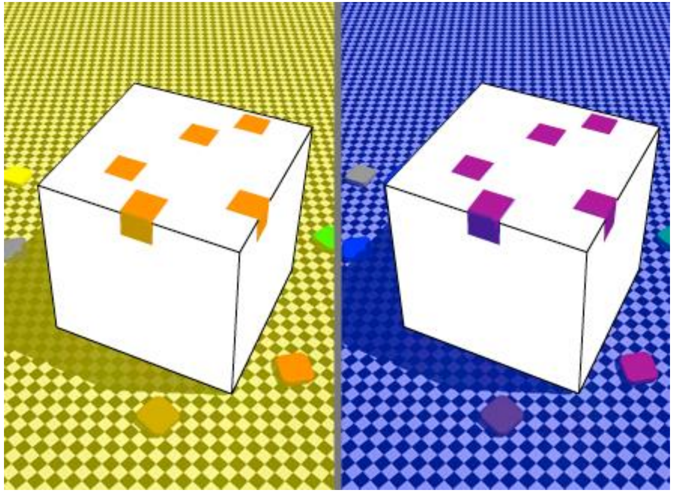


http://www.purveslab.net/seeforyourself/

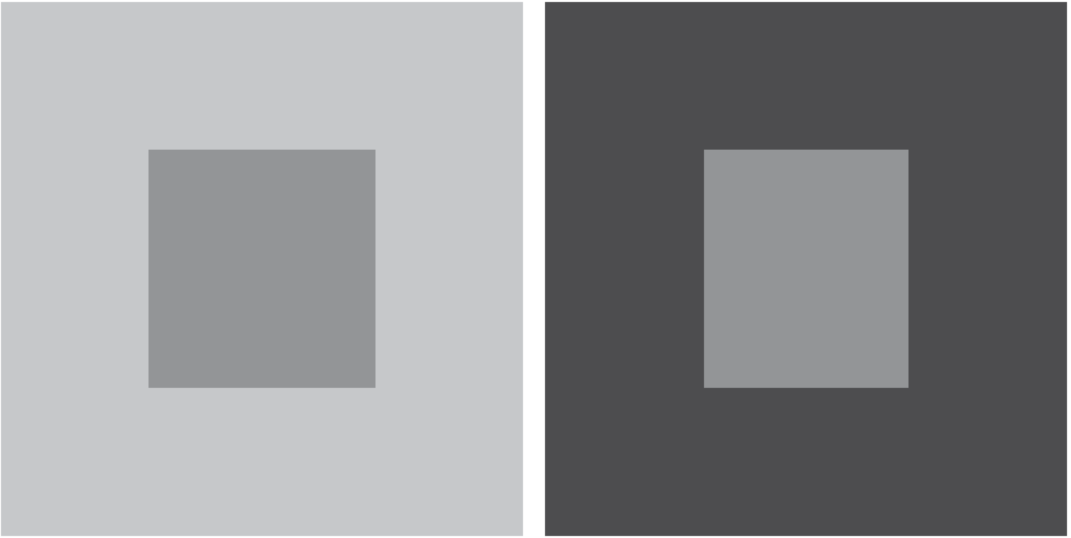
Relative color judgements

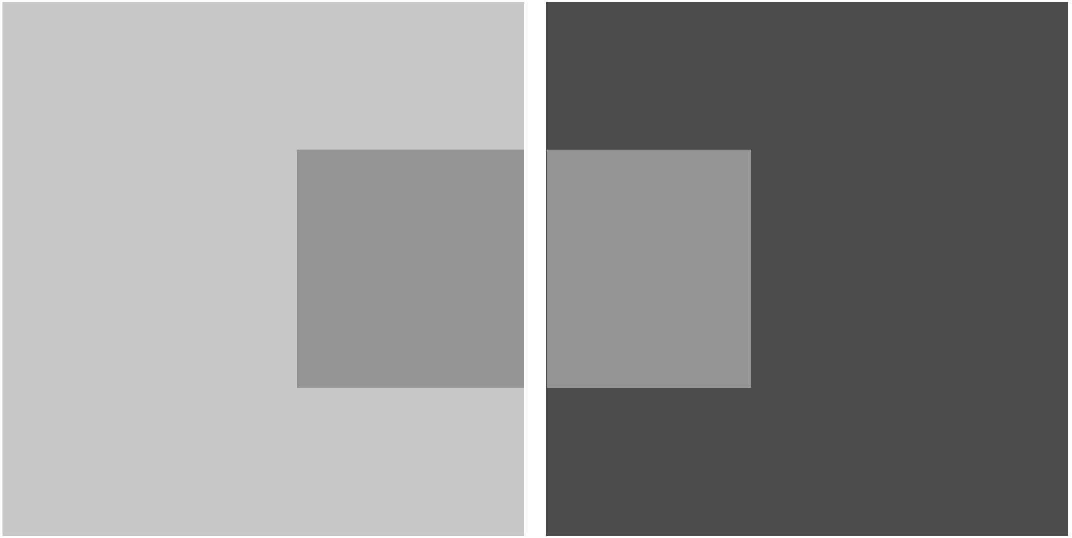
- color constancy across broad range of illumination conditions

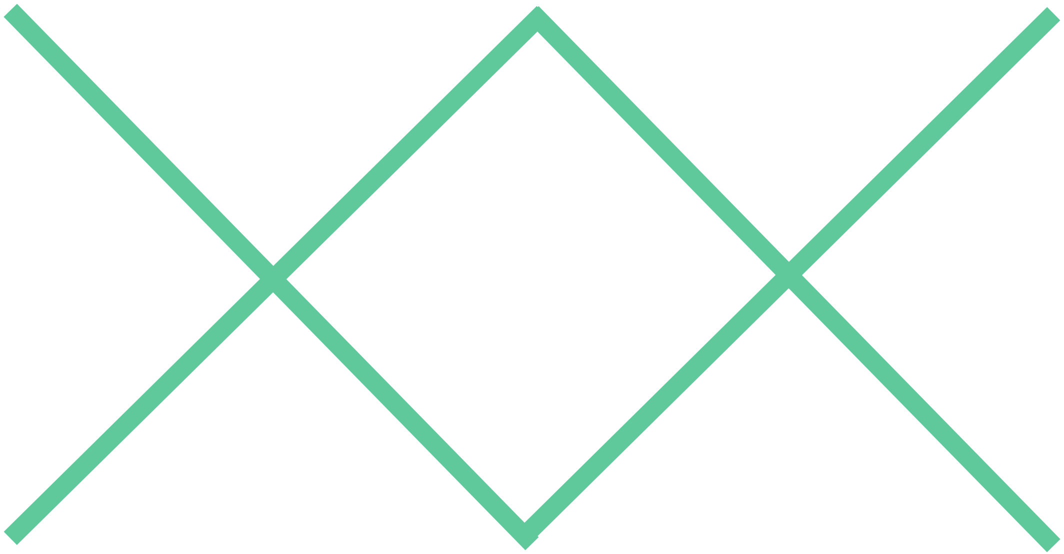


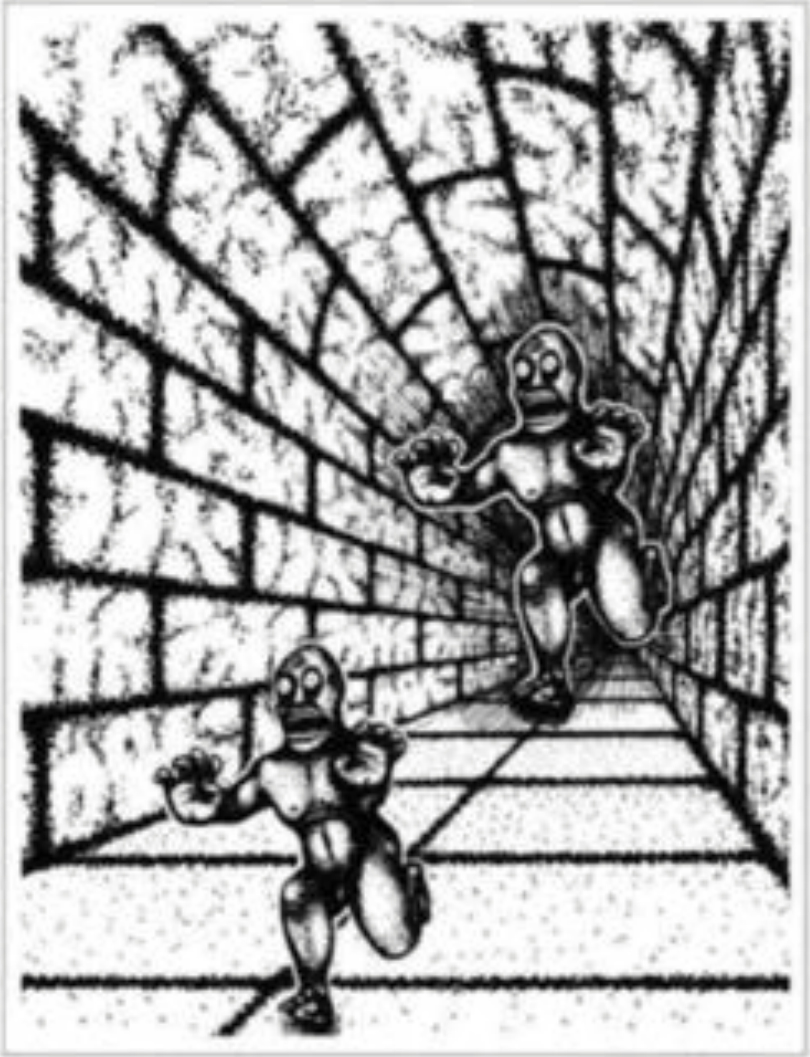


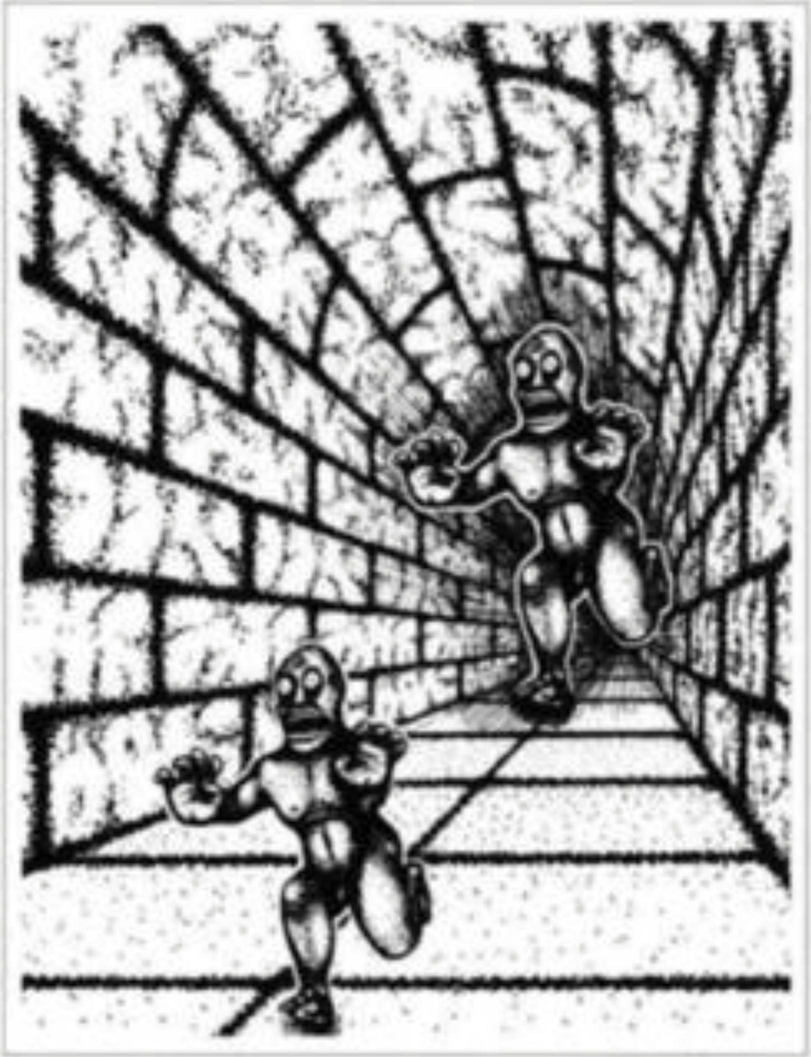
http://www.purveslab.net/seeforyourself/












Cleveland’s three visual operations of pattern perception:

1. Detection
2. Assembly
3. Estimation

2. Assembly

Gestalt psychology of perceptual organisation

- Based on the work of Kurt Koffka, Max Wertheimer, and Wolfgang Köhler
- Law of Prägnanz (pithiness, goodness)
- Things are organized spontaneously and assumed to be in the simplest configuration
- Perception as organized and structured wholes rather than the sum of their constituent parts
- Emergent, holistic, interdependent, and in context

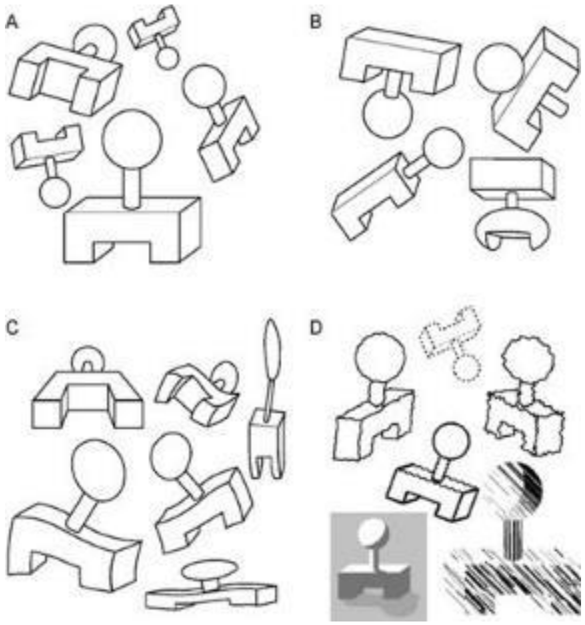
The gestalt laws of perceptual organization

1. Emergence: The mind sees the whole and then the parts. It often sees more than what is specifically stated by its individual parts.

MOSLEEP

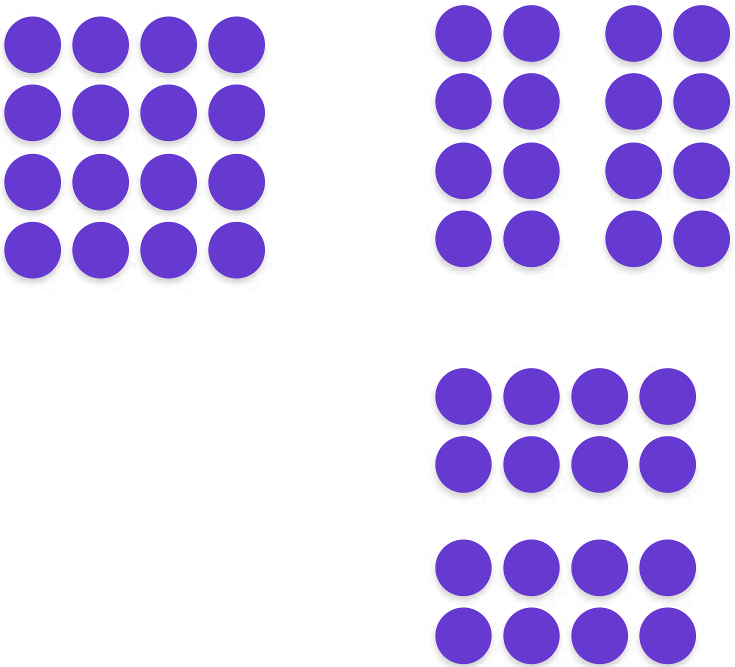
The gestalt laws of perceptual organization

2. Invariance: The mind recognizes simple objects independent of rotation, translation, scale, deformations and lighting



The gestalt laws of perceptual organization

3. Proximity: Elements that are closer together are perceived to be more related than elements that are farther apart

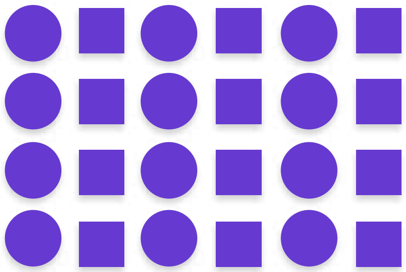


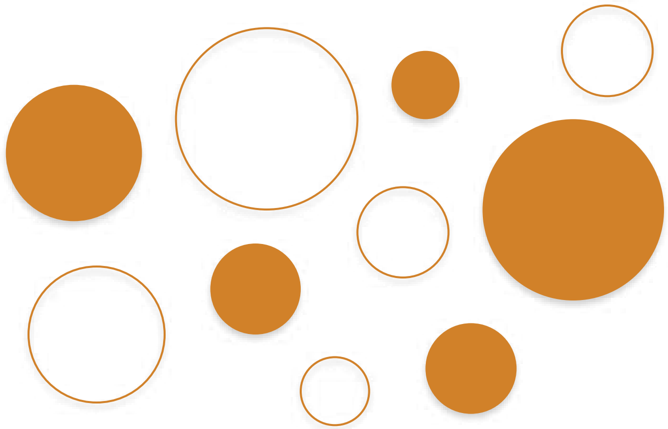
The gestalt laws of perceptual organization

3. Proximity: Elements that are closer together are perceived to be more related than elements that are farther apart

The gestalt laws of perceptual organization

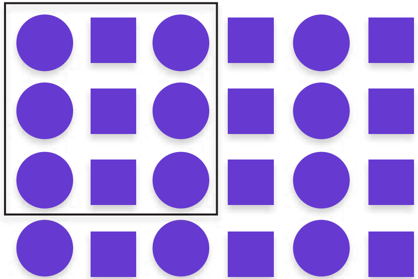
4. Similarity: Elements that are similar are perceived to be more related than elements that are dissimilar





The gestalt laws of perceptual organization

5. Enclosure: Elements that are enclosed by anything are perceived as belonging together

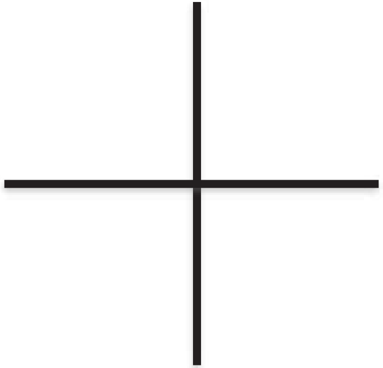


The gestalt laws of perceptual organization

5. Enclosure: Elements that are enclosed by anything are perceived as belonging together

The gestalt laws of perceptual organization

6. Continuity: The mind continues visual, auditory, and kinetic patterns



REUTERS

The gestalt laws of perceptual organization

6. Continuity: The mind continues visual, auditory, and kinetic patterns

The gestalt laws of perceptual organization

7. Closure: The mind perceives a set of individual elements as a single, recognizable pattern

The gestalt laws of perceptual organization

7. Symmetry: The mind perceives objects as symmetrical shapes that form around their center

[ ] { } [ ]

The gestalt laws of perceptual organization

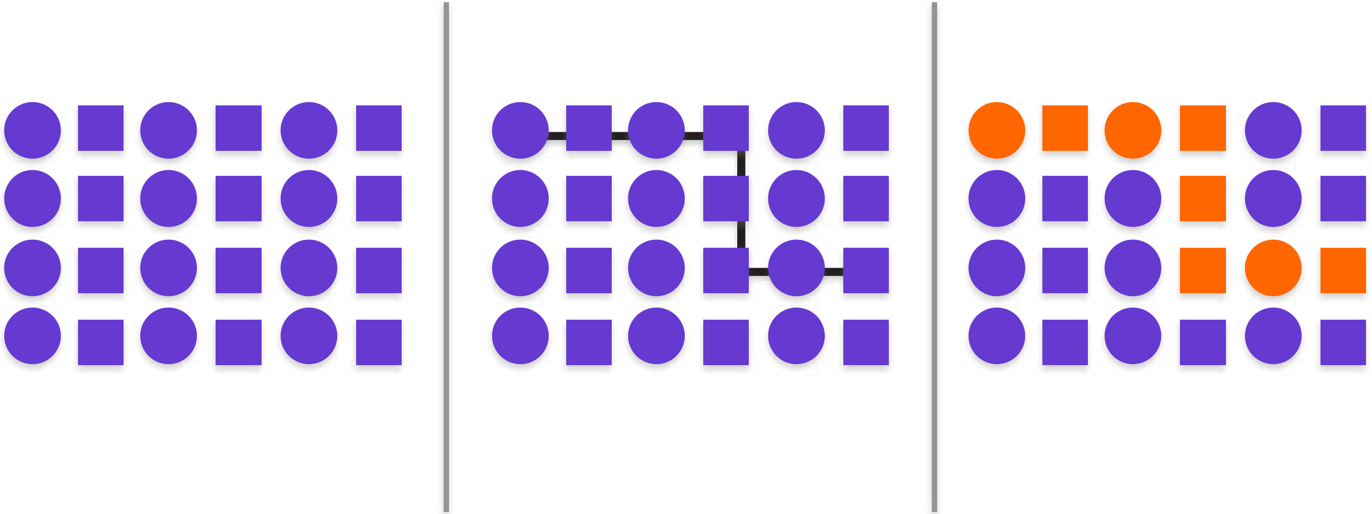
8. Figure-ground: Elements are perceived as either figures (objects of focus) or ground (the rest of the perceptual field)

Formula 1

Hope for African Children Initiative

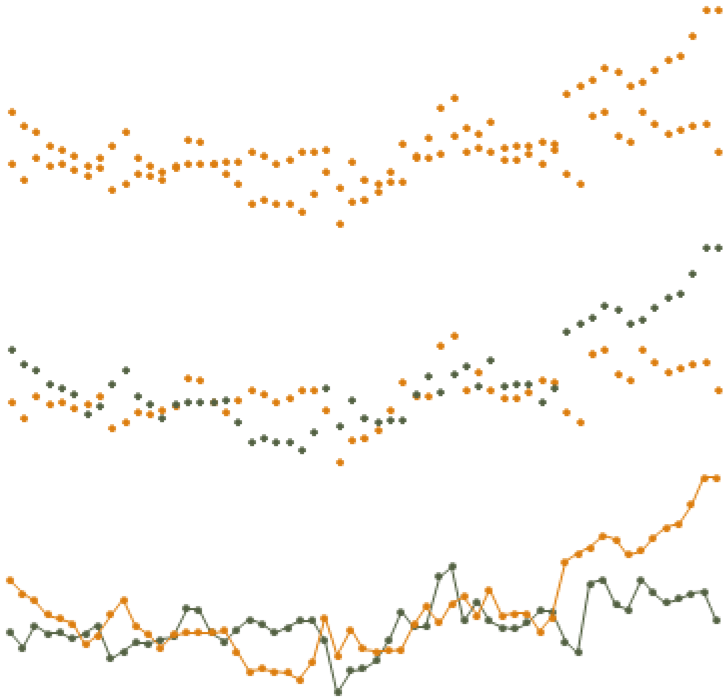
The gestalt laws of perceptual organization

9. Connection: Elements that are connected (e.g. by a line) are perceived as belonging together

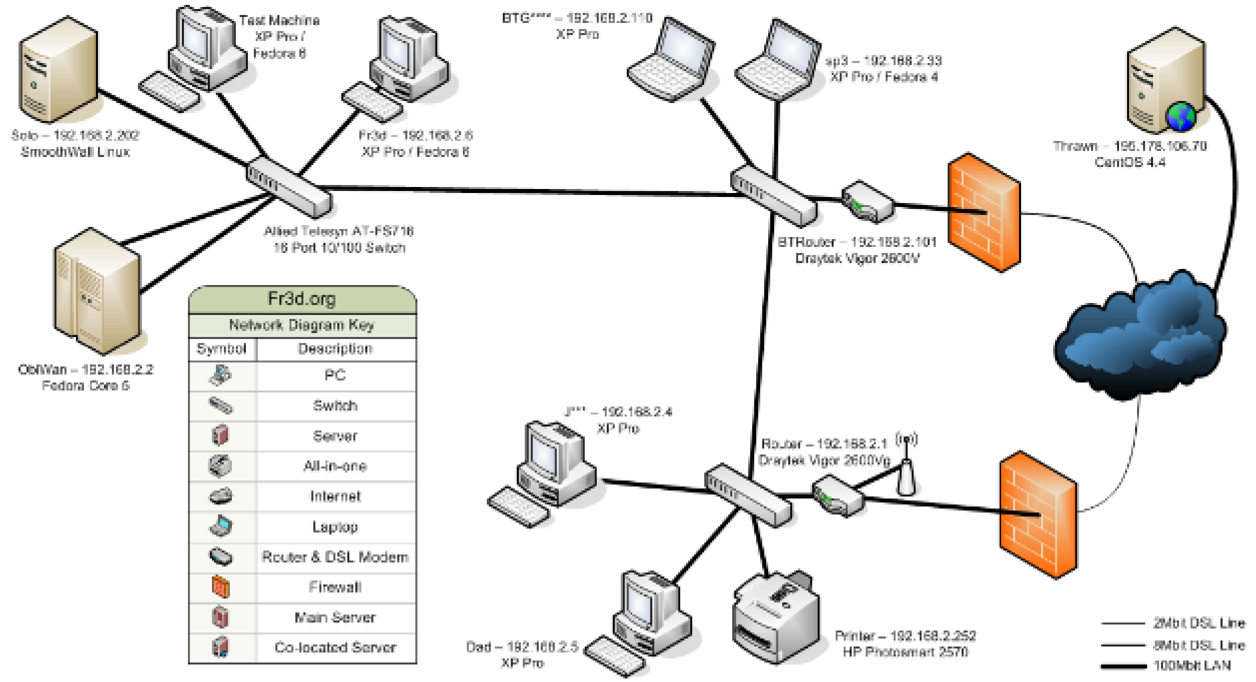


The gestalt laws of perceptual organization

9. Connection: Elements that are connected (e.g. by a line) are perceived as belonging together



9. Connection: Elements that are connected (e.g. by a line) are perceived as belonging together

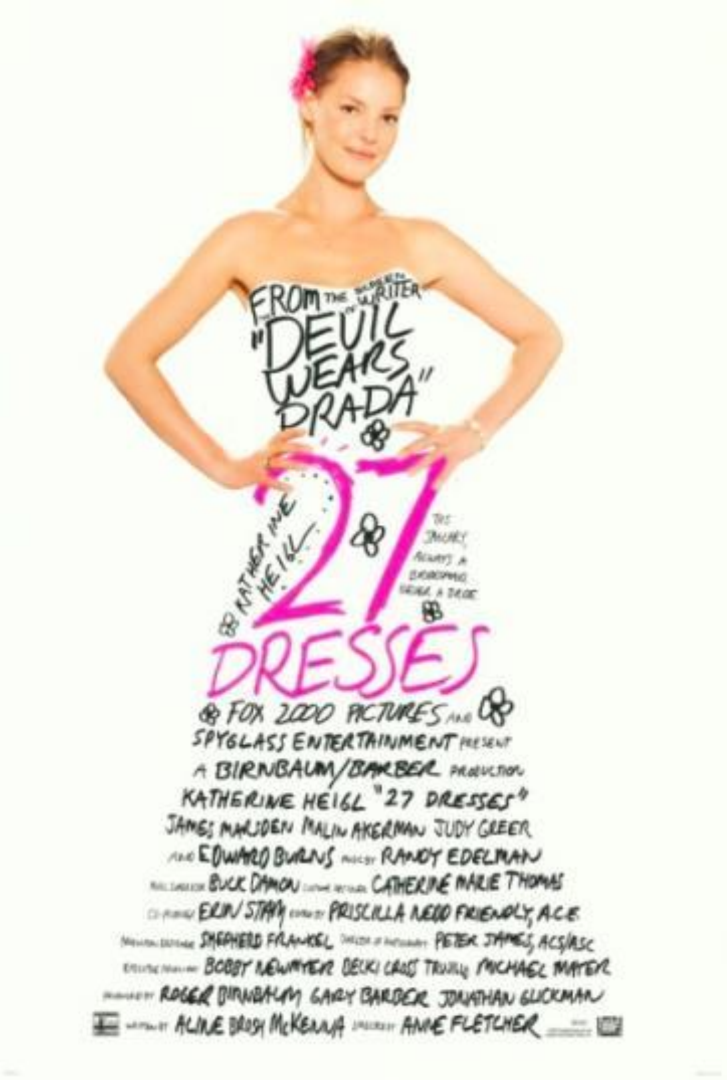


The gestalt laws of perceptual organization

10. Common-fate: Elements that share a common fate (e.g., moving in the same direction) as belonging together

MELBOURNE
2010 CYCLING
UCI Road World Championships

Edward Weston, 1886-1958

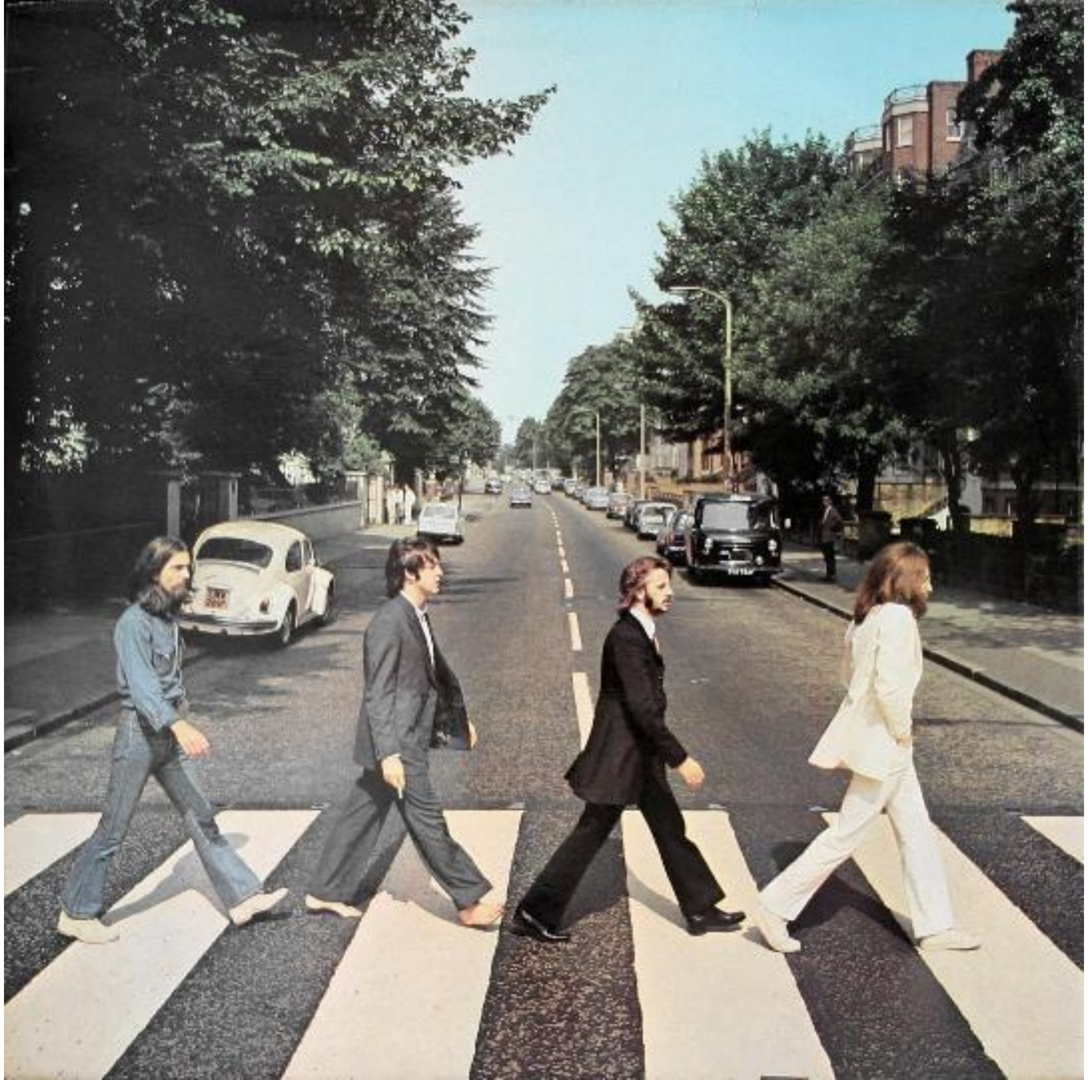

FROM THE WRITER
"DEVIL WEARS PRADA"
KATHERINE HEIGL
27
DRESSES
FOX 2000 PICTURES AND
SPYGLASS ENTERTAINMENT PRESENT
A BIRNBAUM/BARBER PRODUCTION
KATHERINE HEIGL "27 DRESSES"
JAMES MARSDEN MALIN AKERMAN JUDY GREER
AND EDWARD BURNS RANDY EDELMAN
BUCK DAMON CATHERINE MARIE THOMAS
ERIN STAM PRISCILLA NEDD FRIENDLY, A.C.E.
SHEPHERD FRANKEL PETER JAMES, ACS/ASC
BOBBY NEWMYER BECKI CROSS TRUJILLO MICHAEL MAYER
ROGER BIRNBAUM GARY BARBER JONATHAN GLICKMAN
ALINE BROSH McKENNA ANNE FLETCHER


LIMA
MOROCOCHA

The Creation of Adam by Michelangelo, fresco Sistine chapel, 1512

Marc Riboud, 1923-

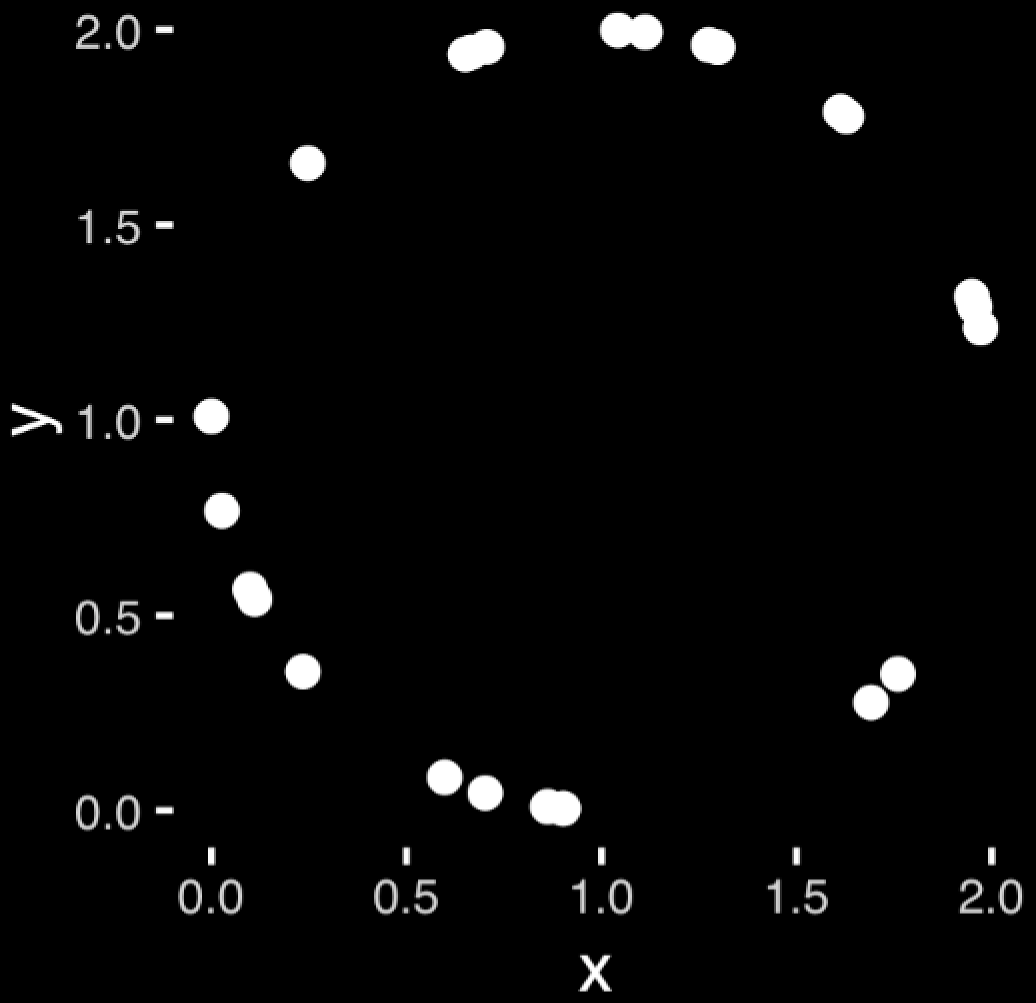
Portrait of Adele Bloch-Bauer.
1907
by Guastav Klimt



Marc Riboud, 1923-

M.C.Escher N° 48/50

eigen druk


2.0
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y
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